Trade and welfare (across local labor markets)*

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Abstract

We derive a sufficient statistic for first-order welfare changes among workers initially living in a region that aggregates changes in wages, prices, unemployment, and regional characteristics (e.g., life expectancy). The framework accounts for worker choices over regions, sectors, and labor force participation and allows arbitrary heterogeneity in initial productivities and non-pecuniary benefits across choices, without specifying how shocks determine observed outcomes or parametrizing a full structural model. We also show how income changes must be measured for welfare accounting in the presence of these choices. Researchers can use the statistic as the outcome in existing research designs that study particular margins of adjustment. Provided the identifying assumptions hold for welfare, these designs identify a shock's differential welfare effects and allow the estimate to be decomposed into contributions from each welfare component. We apply the framework to evaluate the differential welfare effects across U.S. commuting zones of granting China permanent normal trade relations.

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1 Introduction

Recent empirical work has identified substantial differential effects of international trade shocks (across, e.g., regions) on a broad set of adjustment margins, including changes in wages, labor force participation, unemployment, labor allocations across sectors and regions, mortality, crime, and environmental quality. These adjustments naturally raise the question: What are the differential welfare implications of these trade shocks? The standard approach builds and parametrizes a large quantitative trade model featuring a small subset of these margins and strong restrictions on how shocks affect them, introduces the shock, and computes changes in a social welfare function.

In this paper, we take a different approach. We measure welfare changes incorporating each of these margins without fully specifying a structural model, restricting the mechanisms through which shocks affect observed outcomes, or parametrizing model parameters. Constructing this measure requires accounting for how worker choices affect the relationship between observed earnings changes and welfare. Given this sufficient statistic, we can replicate the vast empirical literature's approach, but identify the differential effect of a trade shock on welfare rather than on a particular margin of adjustment. We can additionally decompose this estimate into contributions from each welfare component.

Our contribution is a sufficient statistic in the spirit of Chetty (2009). The statistic applies whether wages are set competitively or not and whether unemployment results from search-and-matching, wage rigidity, or other mechanisms. Our measure aggregates individual welfare changes expressed in income units and is unchanged by alternative utility representations of the same preferences.[1] The statistic is a weighted sum of observed changes in each margin, with weights measured in the initial equilibrium. Hence, it can be constructed from the data the empirical literature already uses. With the statistic as the outcome, research designs studying individual adjustment margins can identify a shock's differential welfare effect, provided their identifying assumptions hold for welfare. We also show that constructing this welfare outcome matters quantitatively. In our application, accounting for workers' optimal choices when measuring the wage component substantially increases the estimated differential welfare loss, and incorporating non-pecuniary unemployment costs and mortality increases it further.

We proceed in two steps. First, we derive a sufficient statistic for workers' welfare changes, to a first-order approximation. Each worker chooses where to live, which determines the consumption prices, employment opportunities, and regional characteristics she

[1] Subsequent work by Baqaee and Burstein (2026) shows that the social welfare functions based on average utility commonly used in quantitative spatial models can yield different welfare comparisons under utility representations that preserve workers' preferences. Our measure aggregates individual equivalent or compensating variations, which coincide to first order.

faces, including (e.g.) life expectancy, environmental quality, and exposure to crime. In that location, each worker chooses whether to participate in the labor force. Participants choose a sector to apply to and are then employed there or unemployed. Wages, prices, unemployment probabilities, and regional characteristics can all change in response to the underlying economic shocks.

Envelope conditions imply that changes in workers' choices over regions, sectors, and labor force participation have no direct first-order effects on welfare. Our welfare sufficient statistic therefore evaluates welfare changes at fixed choices, using workers' initial choices in the baseline formula. For earnings measurement, this requires more than following the same workers over time: their earnings can still change because they move across sectors or regions. These changes can compensate for differences in job amenities, price indices, regional characteristics, or employment prospects and need not represent welfare changes. Labor force exit is an extreme example of this point: earnings can fall dramatically, even though a worker at the participation margin is indifferent between participating and not participating in the workforce. Treating this earnings loss as a welfare loss would ignore the value of non-participation. The wage contribution to welfare therefore requires changes in earnings opportunities within fixed region-sector pairs.

Workers' welfare also depends on the risk of involuntary unemployment. Changes in this risk affect both expected income and expected non-pecuniary benefits. We therefore measure changes in unemployment probabilities separately from changes in earnings conditional on employment, valuing the former using the income and non-pecuniary differences between employment and unemployment. Combining these inputs with changes in prices, monetary valuations of changes in regional characteristics, and initial allocations yields our sufficient statistic.

Second, we show how to implement the approach. We measure welfare changes between 2000 and 2007 for workers initially living in each U.S. commuting zone (CZ), regardless of where they live or work thereafter. We express each welfare component as a share of these workers' initial annual real income. We measure changes in productivity-adjusted wages per efficiency unit from the earnings changes of workers employed in the same CZ-sector pair in both periods, using the Longitudinal Employer-Household Dynamics. We measure local inflation using state-level consumer price indices where available and the national index elsewhere. We measure changes in CZ unemployment rates using the 2000 census and the 2006-2008 American Community Survey. Combining these inputs with initial income shares yields our simpler measure of welfare changes, based on the real wage and pecuniary unemployment components alone. These components reflect all shocks over the period. The more complete measure also incorporates non-pecuniary costs of unemployment and changes in mortality. Specifically, we use existing estimates based on self-reported well-

being to value the non-pecuniary costs of unemployment at twice its income costs. A similar ratio appears in the optimal unemployment insurance analysis of Landais et al. (2018), who value non-pecuniary unemployment costs at roughly twice the welfare loss from reduced consumption. To value changes in mortality, we combine published estimates of excess mortality with the value of a statistical life adopted by the U.S. Department of Health and Human Services.

We use our simpler and more complete welfare measures to study a trade policy change widely examined in the literature: granting China permanent normal trade relations (PNTR). Using a standard differences-in-differences design, we estimate its differential welfare effects across CZs. The application shows why each step in moving from nominal earnings effects to welfare effects matters quantitatively. Our welfare measure requires evaluating changes in earnings opportunities at workers' initial choices. Doing so yields a substantially larger differential loss through the wage component than would be inferred from tracking their earnings as they move across sectors and regions, a common approach in empirical work. One possible explanation is that workers in more exposed CZs move to jobs or regions where their non-pecuniary benefits are lower, with earnings gains from these moves compensating for the loss of those benefits. Moving from nominal to real wages also matters quantitatively: more exposed CZs experience relative price declines, which provide insurance by partly offsetting their nominal wage losses. Combining this real wage component with the income costs of unemployment yields our simpler welfare measure. A CZ at the 90th percentile of exposure experiences a smaller welfare increase than a CZ at the 10th percentile, with a difference equivalent to 1.9% of initial income. Real wage changes account for about 90% of this differential effect. Finally, we value welfare costs not captured by income changes alone. Including non-pecuniary unemployment costs and mortality in our more complete measure increases the differential loss by about 27%, from 1.9% to 2.4% of initial income. These costs also change the composition of the estimated loss: the real wage share falls from about 90% to 71%, with unemployment accounting for 23% and mortality for 6%.

Literature. Our paper builds on empirical work studying the effects of trade shocks on wages, unemployment, labor force participation, the allocation of workers across sectors and regions, mortality, crime, and pollution; see, e.g., Topalova (2010), Kovak (2013), Autor et al. (2013), Autor et al. (2014), Autor et al. (2015), Dix-Carneiro and Kovak (2017), Dix-Carneiro et al. (2018), McCaig and Pavcnik (2018), Dix-Carneiro and Kovak (2019), Bombardini and Li (2020), and Finkelstein et al. (2026); our application uses the same trade shock as Pierce et al. (2024), who study the effects of PNTR on sectoral and regional wages. We combine these adjustments into a single welfare measure without specifying how trade shocks affect them.

Our sufficient statistic approach builds on Chetty (2009) and its application to trade in

Arkolakis et al. (2012). It is most closely related to Porto (2006), Atkin et al. (2018), and Borusyak and Jaravel (2021), who use first-order approximations to measure the distributional welfare effects of trade. We develop the income measurement needed for welfare accounting when workers choose their sectors, locations, and labor force participation, trading off earnings against non-pecuniary benefits. These choices require distinguishing changes in earnings opportunities within fixed region-sector pairs from earnings changes associated with reallocation, and distinguishing unemployment from chosen labor force exit. Our sufficient statistic combines these income components with non-pecuniary unemployment costs and changes in regional characteristics.[2] Constructing the income components of welfare also connects our approach to the Roy-model literature on selection, which distinguishes changes in wages per efficiency unit from changes in workforce composition; see, e.g., Dix-Carneiro (2014) and Adão (2015).

Our approach complements quantitative models of trade and local labor markets; see, e.g., Adão (2015), Caliendo et al. (2018), Kim and Vogel (2021), Galle et al. (2023), Adão et al. (Forthcoming), and Rodriguez-Clare et al. (2026). Relative to this literature, we measure welfare without specifying the mechanisms through which shocks affect observed outcomes, which allows our approach to accommodate a broader range of economic responses while reducing the scope for model misspecification and parametrization error. The resulting welfare measure can be constructed from observed data and monetary valuations and used as the outcome in empirical regressions. Our analysis also has three limitations relative to the structural approach. First, our formula omits higher-order effects for finite changes, including gains from switching choices. To assess sensitivity to this approximation, we provide sufficient statistics that are first-order approximations to lower and upper bounds on consumption-equivalent welfare changes. These statistics can be constructed without introducing a full structural model. Second, predicting the differential welfare effects of counterfactual policies requires additional assumptions about how wages, prices, employment probabilities, and regional characteristics would respond. Quantitative models are well suited to these questions. Third, our regressions identify differential effects of PNTR across CZs and leave its common effect unidentified, even though our welfare measure itself reflects the impact of all shocks over the period, including changes common to all CZs. Quantitative models can help address this 'missing intercept.'

[2] Like Jones and Klenow (2016), we combine monetary and non-pecuniary effects. We use workers' optimal choices to account for earnings and amenity tradeoffs without estimating their valuations of every job and location. We also build on sufficient statistic analyses of unemployment insurance in Baily (1978) and Chetty (2006) and the related literature.

2 A theory of measurement

2.1 Worker choices

There is a continuum of workers, $\omega \in \Omega$. Each worker chooses a region $r \in \{1, ..., R\}$ and a sector $s \in \{0, ..., S\}$ in which to apply for work. A sector can represent an industry, occupation, firm, or their intersection. Sector $s = 0$ is home production, or labor force non-participation. A worker who applies to sector $s \geq 1$ is employed with probability E_{rs} and unemployed otherwise. A non-participant works in home production with certainty, so $E_{r0} = 1$. We use $\mathcal{J} \equiv \{u, 0, 1, ..., S\}$ to denote all possible "jobs," including unemployment, u, and home production.

Worker ω has productivity $\varepsilon_{rj}^{\omega} > 0$ and non-pecuniary benefits η_{rj}^{ω} in each region-job pair. Income per efficiency unit is $\widetilde{w}_{rj}$, so nominal income is $\varepsilon_{rj}^{\omega}\widetilde{w}_{rj}$. This notation also covers unemployment income and the value of home production. Workers have common homothetic preferences over consumption goods. Given the regional price index P_r, real consumption is

$$C_{rj}^{\omega} = \frac{\varepsilon_{rj}^{\omega}\widetilde{w}_{rj}}{P_r}.$$

Utility is the sum of consumption, job amenities, and the amenity value of the region:

$$U_{rj}^{\omega} = C_{rj}^{\omega} + \eta_{rj}^{\omega} + \sum_{m=1}^{M} \phi_m^{\omega}(z_{rm}).$$

Here z_{rm} denotes regional characteristic m, such as life expectancy, environmental quality, or exposure to crime. Each continuously differentiable function ϕ_m^{ω} gives worker ω's amenity value of that characteristic in units of annual real consumption. We normalize each worker's coefficient on consumption to one. This fixes the units of utility without restricting valuations of job amenities and regional characteristics relative to consumption, which can differ arbitrarily across workers. Workers are risk neutral over income lotteries.[3] A worker who chooses (r, s) therefore obtains expected utility

$$\mathbb{E}[U^{\omega}(r, s)] = E_{rs}U_{rs}^{\omega} + (1 - E_{rs})U_{ru}^{\omega}.$$

Each worker knows the returns, employment probabilities, and regional characteristics associated with every region-sector pair and chooses a pair that maximizes this expression. We allow arbitrary heterogeneity in productivity and amenities across workers and choices. The

[3] Online Appendix Section A.4 provides formulas for the welfare outcome and its components under constant relative risk aversion. Under additional restrictions, these use the same data as the baseline, together with a choice of the risk-aversion parameter.

same analysis applies within observable worker groups, allowing their wages, employment probabilities, and price indices to differ.

We consider changes in wages, prices, employment probabilities, productivity, and regional characteristics, holding the job amenities η_{rj}^{ω} fixed. Productivity changes within a region-job pair are proportional and common across workers: $d \log \varepsilon_{rj}^{\omega} = d \log \varepsilon_{rj}$. We write $d \log w_{rj} \equiv d \log \widetilde{w}_{rj} + d \log \varepsilon_{rj}$ for the change in productivity-adjusted wages per efficiency unit. Workers take these variables as given when choosing where to live and work. Their equilibrium determination is left unspecified.

2.2 A sufficient statistic for welfare changes

Let Ω_r be the set of workers initially living in region r, and let $\Omega_{rs} \subseteq \Omega_r$ be those initially applying to sector s. For worker ω, equivalent variation, Y^{ω}, is the fixed annual nominal transfer that, under initial conditions and before the employment lottery, makes the worker indifferent between the initial and terminal environments. The transfer is independent of the worker's choice and employment outcome. We sum these transfers over the initial residents of r and divide by their initial expected annual nominal income, $P_r RI_r$, where P_r is the initial price index and RI_r is initial expected annual real income:

$$W_r \equiv \frac{1}{P_r RI_r} \int_{\Omega_r} Y^{\omega} d\omega.$$

Thus, $W_r = 0.01$ denotes a welfare gain equivalent to 1% of initial annual income.[4] The comparison follows the initial residents wherever they subsequently live or work, including any productivity gains as they age. This is a static comparison of annual welfare.

Suppose the optimal region-sector choice is unique for almost every worker in the initial equilibrium. Applying the envelope theorem to workers' choices and aggregating their welfare changes yields, to first order,

$$\begin{aligned} W_r \approx & \sum_{s=0}^{S} \frac{RI_{rs}^{e}}{RI_r} d \log \left(\frac{w_{rs}}{P_r} \right) + \frac{RI_r^{u}}{RI_r} d \log \left(\frac{w_{ru}}{P_r} \right) \\ & + \sum_{s=1}^{S} \frac{RI_{rs}^{e} - \frac{E_{rs}}{1-E_{rs}} RI_{rs}^{u}}{RI_r} d \log E_{rs} + \sum_{s=1}^{S} \frac{\eta_{rs}^{e} - \frac{E_{rs}}{1-E_{rs}} \eta_{rs}^{u}}{RI_r} d \log E_{rs} \\ & + \sum_{m=1}^{M} \frac{\Phi_{rm}}{RI_r} dz_{rm}. \end{aligned} \tag{1}$$

[4] To first order, this measure also equals aggregate compensating variation divided by initial annual nominal income when almost every worker has a unique initial choice. We refer to it as the welfare change throughout.

All quantities entering the weights are evaluated in the initial equilibrium. For workers initially applying to (r, s), expected total real income from employment and from unemployment are given by

$$RI^e_{rs} \equiv E_{rs} \int_{\Omega_{rs}} C^\omega_{rs} d\omega \qquad \text{and} \qquad RI^u_{rs} \equiv (1 - E_{rs}) \int_{\Omega_{rs}} C^\omega_{ru} d\omega,$$

respectively. Employment lotteries average out across the continuum of workers, so these expected totals equal realized income totals. We can therefore use aggregate incomes to construct these weights without estimating individual productivities. For these same applicants, expected total non-pecuniary benefits received in employment and unemployment are

$$\eta^e_{rs} \equiv E_{rs} \int_{\Omega_{rs}} \eta^\omega_{rs} d\omega \qquad \text{and} \qquad \eta^u_{rs} \equiv (1 - E_{rs}) \int_{\Omega_{rs}} \eta^\omega_{ru} d\omega,$$

respectively. Summing across sectors gives expected total unemployment income and total income,

$$RI^u_r \equiv \sum_{s=1}^{S} RI^u_{rs} \qquad \text{and} \qquad RI_r \equiv \sum_{s=0}^{S} RI^e_{rs} + RI^u_r,$$

respectively. Finally,

$$\Phi_{rm} \equiv \int_{\Omega_r} \frac{d\phi^\omega_m(z_{rm})}{dz_{rm}} d\omega$$

sums initial residents' marginal valuations of characteristic m in annual real income units. Each valuation holds the other characteristics fixed. Online Appendix Section A.1 provides the derivation. Online Appendix Section A.2 also gives a first-order formula for a consumption-equivalent welfare measure using terminal choices, allowing a positive mass of workers to be initially indifferent between optimal choices.

Intuition. To understand equation (1), we introduce the margins of adjustment one at a time. First suppose each region has a single job, workers cannot move, and regional characteristics do not change. Welfare changes then equal region-specific changes in log real wages. Second, introduce many jobs and changing regional characteristics, while holding workers in their initial allocations and keeping employment probabilities fixed. Welfare changes then equal changes in log real income from employment, unemployment, and home production, weighted by initial income shares, as in the first line of equation (1), together with the valuation of changes in regional characteristics, as in the third line.

Third, allow workers to change their region, sector, or labor force participation. The envelope condition and our assumption that the mass of initially indifferent workers is zero imply that these changes have no separate first-order welfare effect: workers at the switching margin are indifferent in expected utility, with differences in earnings offset by differences

in prices, amenities, or employment prospects. We can therefore evaluate first-order welfare changes at workers' initial choices.

Finally, allow employment probabilities to change. Like regional characteristics, these probabilities describe conditions workers face given their choices. Workers choose their region and sector, but not whether they become unemployed. Changes in employment probabilities therefore affect welfare at fixed choices through both expected income and expected non-pecuniary benefits. The second line values these changes using the differences in income and non-pecuniary benefits between employment and unemployment.

Implications for measurement. Equation (1) measures first-order welfare changes using initial allocations and incomes; observed changes in wages, prices, employment probabilities, and regional characteristics; and monetary valuations of non-pecuniary effects. Three implications for measurement deserve emphasis.

First, applying the welfare formula does not require parametrizing the full distribution of worker productivities and preferences or specifying production, wage setting, or the sources of unemployment. These benefits come with two limitations. One is that the formula omits higher-order effects for finite changes, including gains from switching choices. A fully specified structural model can account for these effects, although its additional assumptions and parameters create scope for model misspecification and parametrization errors that our approach helps avoid. Our sufficient statistic approach also provides a way to assess sensitivity to this approximation using observed data, without adding model structure. Online Appendix Section A.3 shows that evaluating the exact change in expected utility at workers' initial choices gives a lower bound on a consumption-equivalent welfare measure, while evaluating it at their terminal choices gives an upper bound. Equation (1) and the terminal-choice formula in equation (A.1) provide first-order approximations of these bounds. Both can be constructed using observed allocations and outcomes, together with the required monetary valuations of regional characteristics. Comparing the two provides one way to assess the sensitivity of our welfare estimates to the first-order approximation. The other limitation is that our approach values observed changes without predicting how shocks produce them. Using it to predict the welfare effects of policies therefore requires additional assumptions about how wages, prices, employment probabilities, and regional characteristics respond.

Second, measuring the wage component requires going beyond accounting for changes in workforce composition. Average earnings within a region-sector pair can change because the workers employed there change. Following the same initial workers over time holds the group fixed, but their earnings can also change because they move across sectors or regions. Earnings changes associated with these moves can compensate for differences in amenities or employment prospects and need not represent welfare changes. We therefore need to

measure changes in earnings opportunities holding both the worker and the region-sector pair fixed. In the baseline formula, this means measuring how earnings would have changed for the initial workers had they remained employed in their initial pair. Section 3.1 describes our approach to estimating these changes.

Third, changes in employment probabilities must be measured separately from changes in labor force participation. Tracking earnings alone does not distinguish their welfare implications: both unemployment and non-participation can appear as lost earnings. In our framework, changes in participation have no separate first-order welfare effect, while changes in employment probabilities affect workers who need not be indifferent between employment and unemployment. Valuing these probability changes requires the income and non-pecuniary differences between employment and unemployment for the same initial applicants. Similarly, changes in regional characteristics require initial residents' marginal willingness to pay. These particular valuations are required even though we need not estimate the full distribution of worker preferences.

2.3 Imposing an additional restriction

One restriction simplifies implementation. Employment probabilities are common across sectors within a region: $E_{rs} = E_r$ for $s \geq 1$. We impose this restriction because unemployment data do not identify the sector to which a worker applies. Under this restriction, each applicant faces the same change in employment probability, so only the regional average non-pecuniary unemployment cost enters the welfare formula. Expressing initial incomes and valuations in nominal units, equation (1) then becomes

$$W_r \approx \underbrace{\sum_{j \in \mathcal{J}} \frac{I_{rj}}{I_r} d \log \left(\frac{w_{rj}}{P_r} \right)}_{\text{real wage}} + \underbrace{(1 + \eta_r) \frac{I_r^e}{I_r} \left[1 - \frac{\overline{I}_r^u}{\overline{I}_r^e} \right] d \log E_r}_{\text{unemployment}} + \underbrace{\sum_{m=1}^{M} \frac{N_r \overline{\phi}_{m,r}}{I_r} dz_{rm}}_{\text{regional characteristics}} . \tag{2}$$

Here I_{rj} is initial annual nominal income in job j, $I_r \equiv \sum_{j \in \mathcal{J}} I_{rj}$ is total income, and $I_r^e \equiv \sum_{s=1}^{S} I_{rs}$ is income earned in employment. Average incomes among employed and unemployed workers are $\overline{I}_r^e$ and $\overline{I}_r^u$, so their ratio is the unemployment replacement rate. The non-pecuniary cost of unemployment relative to the income loss is $\eta_r \equiv P_r \widetilde{\eta}_r / (\overline{I}_r^e - \overline{I}_r^u)$, where

$$\widetilde{\eta}_r \equiv \frac{\sum_{s=1}^{S} \int_{\Omega_{rs}} (\eta_{rs}^{\omega} - \eta_{ru}^{\omega}) d\omega}{\sum_{s=1}^{S} |\Omega_{rs}|}$$

is the average difference between non-pecuniary benefits in employment and unemployment among all initial applicants in region r. Finally, $N_r \equiv |\Omega_r|$ and $\overline{\phi}_{m,r} \equiv P_r \Phi_{rm} / N_r$ is the

average marginal valuation of characteristic m in annual nominal income units. All these quantities are measured in the initial equilibrium.

The real wage component averages changes in log real wages across jobs using initial income shares. The unemployment component weights changes in employment probabilities, $d \log E_r$, by employed workers' initial income share, I_r^e / I_r. The replacement rate, $\overline{I}_r^u / \overline{I}_r^e$, reduces unemployment's income cost, while $1 + \eta_r$ incorporates its non-pecuniary cost. The final term values changes in regional characteristics relative to initial income.

2.4 Using welfare changes as an empirical outcome

Equations (1) and (2) approximate welfare changes from all shocks over the period, without attributing observed changes to particular shocks or specifying how those shocks affect outcomes. Researchers can use this first-order welfare measure as the dependent variable to identify a shock's effect, provided the research design's identifying assumptions hold for the combined outcome.

The additive formula also permits a decomposition. Applying the same linear regression to each weighted component, with the same observations, controls, and estimation weights, yields coefficients that sum to the total welfare coefficient. This decomposes the estimated effect without specifying how the components affect one another.

3 The welfare effects of granting China PNTR

We first measure welfare changes between 2000 and 2007 for workers initially living in each U.S. commuting zone (CZ). These changes reflect all shocks over the period. We then use this measure to estimate and decompose the differential welfare effects of granting China permanent normal trade relations (PNTR).

3.1 Measuring the welfare components

We use mainland CZs as regions and six-digit 1997 NAICS industries as sectors. Our sample covers approximately 600 CZs in 34 states and 450 sectors.[5] Among the regional characteristics, we focus only on life expectancy, setting $M = 1$. Other characteristics can be included when their changes can be measured and valued in monetary units; see, e.g., Chay and Greenstone (2005) for estimates of willingness to pay for cleaner air. We begin with the income components of equation (2), then add non-pecuniary unemployment costs and

[5] Counts are rounded to comply with U.S. Census Bureau disclosure requirements.

life expectancy. The Online Appendix gives data sources and construction details and Figure D.1 reports the distribution of our simpler measure of welfare changes for the fixed set of workers initially living in each CZ.

Initial income shares. We measure employment income by CZ-sector using wage bills in the 2000 Longitudinal Business Database. We construct unemployment income and the value of home production using census population counts, state unemployment replacement rates, and estimates of time spent in home production and its value. Together these determine initial income and the weights in equation (2).

Wage changes. To measure changes in earnings opportunities at initial choices, we hold both worker composition and the CZ-sector pair fixed. We use the Longitudinal Employer-Household Dynamics (LEHD) to measure average log earnings changes among roughly 15 million workers employed in the same pair in both periods. Under the common proportional productivity changes in Section 2.1, each worker's log earnings change within a pair equals $d \log w_{rs}$, which incorporates both changes in wages per efficiency unit and changes in productivity. Our estimate is unbiased if earnings growth among stayers equals, on average, the earnings growth that all initial workers in the pair would have experienced had they stayed. This identification restriction concerns the growth over time in productivities; productivity levels may differ arbitrarily across workers and region-job pairs. The Online Appendix states this condition formally. We hold real returns to unemployment and home production fixed: $d \log(w_{rj}/P_r) = 0$ for $j \in \{u, 0\}$.

Prices and unemployment. We measure inflation using state-level consumer price indices from Hazell et al. (2022), applying the national index where state data are unavailable. The state-level indices exclude shelter and therefore omit local changes in housing costs.[6] These indices capture variation across states, leaving within-state differences unmeasured. We measure E_r, the employed share of the labor force (i.e., one minus the unemployment rate), using the 2000 census and the 2006-2008 American Community Survey.[7] Using employment relative to the labor force distinguishes unemployment from non-participation, which a measure based only on earnings does not do.

Non-pecuniary unemployment costs. The simpler welfare measure sets $\eta_r = 0$ and includes only the income costs of unemployment. Our more complete measure also includes its non-pecuniary costs. Experimental evidence supports a positive non-pecuniary value of employment: Hussam et al. (2022) find that employment improves refugees' well-being more than

[6] The indices exclude rent of primary residence and owners' equivalent rent. Incorporating housing costs requires local rent changes, housing expenditure shares, and assumptions about housing ownership among the initial workers.

[7] We cannot track unemployment rates for fixed sets of workers within regions: the census and ACS are not panels, and LEHD does not record unemployment.

an equivalent cash transfer, and that many participants are willing to forgo cash payments to continue working without pay. Knabe and Ratzel (2011) estimate that maintaining self-reported well-being after unemployment requires a transfer approximately three times the income loss. We therefore value non-pecuniary costs at twice the income loss, setting $\eta_r = 2$ and tripling the unemployment component. In their analysis of optimal unemployment insurance, Landais et al. (2018) use a similar calibration: non-pecuniary unemployment costs are nearly twice the welfare loss from reduced consumption. Larger non-pecuniary costs relative to income losses are reported by Winkelmann and Winkelmann (1998) and Blanchflower and Oswald (2004).

Life expectancy. The simpler measure omits changes in mortality. Because mortality is not observed in our data, the more complete measure uses published estimates of PNTR's differential mortality effect. Pierce and Schott (2020) estimate that an interquartile-range increase in county exposure, 8.3 percentage points, raises annual drug-overdose mortality by 2 to 3 deaths per 100,000; we use the midpoint. Their exposure measure averages NTR gaps over all industries, assigning zero to industries not subject to tariffs, whereas ours averages over manufacturing sectors with a defined gap. We approximate the conversion using the income-weighted mean manufacturing employment share across our CZs in 1980, 0.23. This implies about 7 additional deaths per 100,000 per year per unit of NTRG_r. We value this increase using the U.S. Department of Health and Human Services value of a statistical life (U.S. Department of Health and Human Services, 2026) and divide by our measure of average annual income per person. This adds $-0.018\,\text{NTRG}_r$ to each CZ's welfare outcome and contributes -0.018 to PNTR's estimated welfare coefficient.

This component omits mortality changes common to all CZs and differential changes due to other shocks. These omissions limit our measurement of total welfare changes, but need not bias our estimate of PNTR's differential welfare effect: we combine its effect on the observed components, identified by our research design, with its published mortality effect. Direct mortality data would allow us to value observed changes, including those omitted here. The Online Appendix gives the calculation and alternative valuations.

Risk aversion. Under the restrictions in Online Appendix Section A.4, the risk-aversion extension uses these same inputs. For a chosen risk-aversion parameter, its formulas adjust component weights and recalculate the non-pecuniary unemployment valuation using the same compensation evidence. The adjusted welfare outcome and its components can be used in the same regressions.

3.2 The trade shock and research design

Following Pierce and Schott (2016) and Handley and Limão (2017), we study the removal of uncertainty about U.S. tariffs on Chinese goods. China received normal trade relations (NTR) tariff treatment every year from 1980 to 2001, but annual renewal was required to avoid higher non-NTR tariffs. Permanent normal trade relations (PNTR), granted in 2000 and effective at the end of 2001, removed this risk. We measure exposure using the sectoral gap between non-NTR and NTR tariff rates in 1990, denoted by NTR gap_s.

We measure CZ exposure as

$$\text{NTRG}_r \equiv \sum_{s=1}^{S} \omega_{rs} \text{NTR gap}_s,$$

where ω_{rs} is the sector's share of CZ employment in the 1980 Longitudinal Business Database. We normalize shares to sum to one over manufacturing sectors with a defined NTR gap, addressing the incomplete-shares issue discussed by Borusyak et al. (2022). Online Appendix Figure D.2 reports the distribution of CZ exposure, weighted by initial employment income.

We estimate

$$W_r = \alpha + \beta \, \text{NTRG}_r + \mathbf{X}_r' \gamma + \epsilon_r. \tag{3}$$

We also replace W_r with each component, holding the sample, controls, and weights fixed; the component coefficients sum to the total welfare coefficient. Regressions weight CZs by initial employment income and cluster standard errors by state. Table 1 lists the controls; the labor force and demographic controls follow Autor et al. (2013).

To identify PNTR's differential welfare effect, we assume that, conditional on the controls, changes in the combined welfare outcome would not systematically vary with exposure in the absence of PNTR. As a check on differential trends, we regress welfare changes between 1992 and 2000 on subsequent PNTR exposure. The estimate is statistically indistinguishable from zero and, if anything, indicates rising welfare in more exposed CZs. Online Appendix Table D.4 reports this exercise and the estimates for 2000-2007 for the same sample of six states and approximately 80 CZs covered by the earlier LEHD.

The general welfare measure W_r reflects all shocks over the period, including changes common to all CZs. The regression, however, leaves PNTR's common effect unidentified because the intercept also reflects the common effects of other shocks; this is the standard 'missing intercept' problem.

	(1)	(2)	(3)	(4)	(5)
Panel A: W_r					
NTRG_r	-0.214^{*}	-0.190	-0.183^{**}	-0.243^{***}	-0.248^{***}
State-clustered SE	(0.122)	(0.120)	(0.088)	(0.084)	(0.096)
90th-10th percentile difference	-0.017	-0.015	-0.014	-0.019	-0.019
Panel B: decomposition of W_r					
Real wage:					
NTRG_r	-0.175	-0.162	-0.162^{*}	-0.211^{***}	-0.224^{**}
State-clustered SE	(0.118)	(0.118)	(0.085)	(0.082)	(0.093)
BHJ SE	(0.096)	(0.085)	(0.062)	(0.066)	(0.069)
Unemployment:					
NTRG_r	-0.039^{***}	-0.027^{***}	-0.021^{**}	-0.032^{***}	-0.024^{***}
State-clustered SE	(0.013)	(0.009)	(0.009)	(0.008)	(0.007)
BHJ SE	(0.007)	(0.009)	(0.009)	(0.009)	(0.009)
Observations	600	600	600	600	600
manufacturing share$_{-1}$		✓	✓	✓	✓
missing NTRG sector share$_{-1}$		✓	✓	✓	✓
employment-to-population$_{-1}$		✓	✓	✓	✓
college share$_{-1}$				✓	✓
foreign born share$_{-1}$				✓	✓
female share$_{-1}$				✓	✓
routine occupation share$_{-1}$					✓
average offshorability$_{-1}$					✓
Census division FE			✓	✓	✓

Panel A reports differential welfare effects across five specifications, including real wages and pecuniary unemployment costs. Panel B decomposes these effects. Standard errors are clustered by state; Panel B also reports Borusyak et al. (2022) standard errors. Regressions are weighted by CZ employment income in 2000. Manufacturing share is the share of employment in manufacturing. College and foreign-born shares are population shares; female and routine-occupation shares are employment shares. These controls and average offshorability are measured in 1990 and taken from Autor et al. (2013). Missing NTRG sector share is the share of 1980 LBD employment in sectors without a defined NTR gap. Employment-to-population is measured in the 1980 census. Observation counts are rounded for disclosure avoidance.

Table 1: Differential welfare effects: real wages and pecuniary unemployment costs.

3.3 Welfare effects and their decomposition

The simpler welfare measure. Panel A of Table 1 reports estimates using the simpler measure, which includes only the real wage and pecuniary unemployment components. In the specification with the full set of controls (column 5), the coefficient of -0.248 implies that a CZ at the 90th percentile of exposure experiences a smaller welfare increase than a CZ at the 10th percentile, with a difference equivalent to 1.9% of initial annual income. Comparing columns 1–5 shows that coefficients are broadly stable as we add controls.

Panel B shows that the real productivity-adjusted wage component accounts for about 90% of this differential effect: its coefficient is -0.224, with the pecuniary unemployment component accounting for the remaining -0.024. Each component is statistically significant; the component estimates using Borusyak et al. (2022) standard errors support the same conclusions. We next examine the wage changes underlying this component, the importance of measuring them at initial choices, and the role of regional prices.

We first examine why productivity-adjusted wage growth is lower in more exposed CZs. One possibility is that productivity-adjusted wage growth is similar across CZs within each

Panel A: alternative earnings outcomes			
	Real wage component (1)	Average earnings, employed in both periods (2)	Earnings, initially employed (3)
$NTRG_r$	-0.266^{***}	-0.169^{**}	-0.127^{**}
	(0.085)	(0.080)	(0.061)
90th-10th percentile difference	-0.021	-0.013	-0.010
Observations	600	600	600
Panel B: additional welfare components			
	Income components (1)	+ non-pecuniary unemployment costs (2)	+ life expectancy (3)
$NTRG_r$	-0.248^{***}	-0.296^{***}	-0.314^{***}
	(0.096)	(0.104)	(0.104)
90th-10th percentile difference	-0.019	-0.023	-0.024
Observations	600	600	600
Contributions to the coefficient:			
Real wage	-0.224	-0.224	-0.224
Pecuniary unemployment cost	-0.024	-0.024	-0.024
Non-pecuniary unemployment cost		-0.048	-0.048
Life expectancy			-0.018

All columns use the specification in column 5 of Table 1. Earnings changes in Panel A are measured from LEHD records for 1999Q2 and 2007Q2 for workers aged 18 to 57 in 1999Q2 who hold a single job. Earnings are total quarterly earnings, deflated by the national CPI-U-RS, and changes are converted to a seven-year equivalent. Column 1 uses the real wage component of equation (2) with a national price index rather than the state price indices in Table 1; it is constructed from workers employed in the same CZ-sector pair in both quarters. Column 2 fixes the set of workers employed in the CZ in 1999Q2 who have positive earnings in 2007Q2 in any CZ or sector, and takes the change in their mean log earnings. Column 3 fixes the set of all workers employed in the CZ in 1999Q2, computes mean earnings in levels in each quarter, allowing for zero earnings in 2007Q2, and takes the difference in the logs of these means. In Panel B, column 1 includes real wages and pecuniary unemployment costs. Column 2 adds non-pecuniary unemployment costs with $\eta_r = 2$. Column 3 adds the mortality contribution using Pierce and Schott (2020), U.S. Department of Health and Human Services (2026), and U.S. Bureau of Economic Analysis (2026), taking 2.5 additional deaths per 100,000 per interquartile-range increase in their exposure measure, converted to $NTRG_r$ units using the mean 1980 manufacturing employment share of 0.23 (Online Appendix). Contributions decompose the coefficient; blank cells indicate omitted components. Standard errors are clustered by state and, in Panel B, condition on the valuations and published mortality estimate. Regressions are weighted by CZ employment income in 2000. Observation counts are rounded for disclosure avoidance.

Table 2: Differential effects: additional results.

sector, with more exposed CZs earning larger shares of their initial income in sectors with lower wage growth. Another is that wages grow less within sectors in more exposed CZs. Online Appendix Table D.2 examines the CZ-sector wage changes used to construct our welfare measure. Regressing productivity-adjusted wage growth on separate measures of sector and CZ exposure yields a negative and statistically significant coefficient on CZ exposure, while the coefficient on sector exposure is close to zero. This supports the second explanation: wage growth is lower within sectors in more exposed CZs. Similarly, Pierce et al. (2024) find that workers' earnings following PNTR depend more on their initial location's exposure than their initial industry's. Their measure follows workers' realized earnings across sectors and regions, while ours measures changes in earnings opportunities at initial choices. This qualitative agreement provides a useful check on the regional pattern of wage changes underlying our welfare estimates.

We next examine the quantitative importance of measuring earnings opportunities at workers' initial choices, as the welfare sufficient statistic requires. Panel A of Table 2 applies the specification in column 5 of Table 1 to our real wage component and two alternative

earnings outcomes, using a common national price index to facilitate comparison. Column 1 uses the real wage component of equation (2), constructed from earnings changes among workers employed in the same CZ-sector pair in both periods and aggregated using initial income shares. Columns 2 and 3 follow workers' realized earnings across CZs and sectors and are related to the worker-level earnings outcomes studied by, e.g., Autor et al. (2014) and Pierce et al. (2024).

Column 2 measures earnings changes for workers initially employed in a CZ who are employed in both periods, including those who move across CZs or sectors. We average their individual log earnings changes. Like our wage component, this outcome excludes earnings losses from leaving employment: our welfare measure values unemployment separately and assigns no direct first-order welfare effect to changes in labor force participation. Column 3 measures earnings changes for all workers initially employed in a CZ, including those no longer employed in the second period. Including these workers brings it closer to the standard approach in the literature. For this fixed group, we compute mean earnings in each period, including zero earnings for those no longer employed, and take the difference in the logs of these means. The Online Appendix provides construction details, including differences in weighting and sample coverage across the measures.

Both alternative earnings outcomes yield substantially smaller estimated differential losses than the real wage component: the coefficient is -0.266 in column 1, compared with -0.169 in column 2 and -0.127 in column 3. One possible explanation for the difference between our approach and these alternatives is that workers in more exposed CZs move to jobs or regions where their non-pecuniary benefits are lower, with earnings gains from these moves compensating for the loss of those benefits. Whatever the explanation, using either alternative earnings outcome as the wage component would substantially understate the estimated differential welfare loss through this channel.

Finally, we examine the importance of using regional price indices. The comparisons in Panel A of Table 2 use a common national price index. Accounting for regional price changes reduces the differential loss through the wage component from 2.1% to 1.7% of initial annual income between CZs at the 90th and 10th percentiles of exposure (Online Appendix Table D.1, columns 1 and 2). More exposed CZs experience relative price declines that partly offset their nominal wage losses, providing insurance against PNTR's differential effects. Relatedly, Feler and Senses (2017) document relative declines in housing prices and rents in U.S. commuting zones more exposed to Chinese import competition. Atkin (2024) uses these estimates to illustrate how lower housing costs can partly offset income losses. Our estimates capture the offset to nominal wage losses from measured price changes; local rent declines could provide additional insurance for renters.

The more complete welfare measure. Here, we value welfare costs that income changes

alone do not capture. Panel B of Table 2 fixes the empirical specification and adds the remaining welfare components. Column 1 restates the estimate using the simpler measure from column 5 of Table 1. Valuing non-pecuniary unemployment costs at $\eta_r = 2$ triples the unemployment coefficient (by construction) to -0.072, raising the differential effect from 1.9% to 2.3% of initial income (column 2). Adding the life expectancy component contributes a further loss equivalent to 0.14% of annual income, bringing the differential to 2.4% (column 3).

These additional costs change the composition as well as the size of the estimated welfare effect. Including both raises the differential effect by about 27% and reduces the wage share from about 90% to 71% (Panel B). Unemployment then accounts for 23% and life expectancy for 6%.[8] Real wage changes remain the largest source of the differential welfare loss, but the income costs of unemployment alone substantially understate its welfare importance.

4 Concluding remarks

We provide a sufficient statistic that combines initial allocations, observed adjustments, and valuations of non-pecuniary costs to measure welfare changes to first order. It accounts for worker choices over region of residence, sector of employment, and labor force participation, together with unemployment and regional characteristics. We allow arbitrary differences in initial productivities and non-pecuniary benefits across workers and region-job pairs. Our approach does not restrict how shocks affect wages, prices, employment probabilities, or regional characteristics, and does not require writing or parametrizing a full structural model. Using this measure as an empirical outcome allows a researcher to identify a shock's differential welfare effect, provided the research design's identifying assumptions hold for the combined outcome. In addition, applying the same regression to each component decomposes the estimated effect.

Our PNTR application shows how to construct the measure and why doing so matters. The theory determines both how these components enter welfare and what the income components must measure: changes in earnings opportunities within fixed region-sector pairs, together with changes in unemployment probabilities. Evaluating changes in earnings opportunities at workers' initial choices implies substantially larger differential losses through the wage component than would be inferred from tracking their earnings across sectors and regions. Accounting for non-pecuniary unemployment costs and mortality increases the estimated welfare losses further. These findings demonstrate the quantitative importance of moving from earnings effects to welfare effects. The approach applies to other shocks and to

[8] The Online Appendix reports sensitivity to treating non-participants as unemployed in Table D.3.

other observable regional characteristics, such as pollution and crime, whose welfare effects can be valued in monetary units.

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Online Appendix to "Trade and welfare (across local labor markets)"

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A Theoretical derivations

A.1 Baseline case

Consider worker ω whose initial optimal choice is (r, s). Expected utility is

$$V^\omega = E_{rs}(C^\omega_{rs} + \eta^\omega_{rs}) + (1 - E_{rs})(C^\omega_{ru} + \eta^\omega_{ru}) + \sum_{m=1}^{M} \phi^\omega_m(z_{rm}).$$

Given common proportional productivity changes, as described in Section 2.1 of the main text, $dC^\omega_{rj} = C^\omega_{rj} d\log(w_{rj}/P_r)$. The job amenities η^ω_{rj} are fixed. Almost every worker has a unique initial optimal choice, so the envelope theorem gives the first-order change in maximized expected utility by differentiating at that choice.

Equivalent variation for worker ω, Y^ω, is the fixed annual nominal transfer under initial conditions and before the employment lottery that makes the worker indifferent between the initial and terminal environments. The transfer is independent of the worker's choice

and employment outcome. Its marginal expected utility at the initial choice is $1/P_r$, so

$$\begin{aligned}\frac{Y^\omega}{P_r} \approx{}& E_{rs}C_{rs}^\omega d\log\left(\frac{w_{rs}}{P_r}\right) + (1-E_{rs})C_{ru}^\omega d\log\left(\frac{w_{ru}}{P_r}\right)\\ &+ E_{rs}\left[C_{rs}^\omega - C_{ru}^\omega + \eta_{rs}^\omega - \eta_{ru}^\omega\right] d\log E_{rs}\\ &+ \sum_{m=1}^{M}\frac{d\phi_m^\omega(z_{rm})}{dz_{rm}}dz_{rm}.\end{aligned}$$

All levels on the right-hand side are initial values. The final sum follows from differentiating each ϕ_m^ω with respect to z_{rm}. For a non-participant, $E_{r0} = 1$ and $d\log E_{r0} = 0$.

Integrating Y^ω/P_r over Ω_{rs}, summing over initial sectors, and dividing by RI_r gives equation (1) of the main text. In particular, $E_{rs}\int_{\Omega_{rs}} C_{ru}^\omega d\omega = E_{rs}RI_{rs}^u/(1-E_{rs})$, which accounts for the adjustment to unemployment income in the third term. The same calculation applies to unemployment amenities in the fourth term. Changes in the sets of workers choosing each region-sector pair introduce no additional first-order terms because workers are indifferent between choices at the switching margin.

When employment probabilities are common across sectors within a region, the employment-probability terms can be aggregated across applicants. Valuing incomes and amenities at initial prices gives equation (2) of the main text. In particular, $I_r = P_r RI_r$ and $N_r\overline{\phi}_{m,r} = P_r\Phi_{rm}$ for each m, so each term for regional characteristics is unchanged by expressing its numerator and denominator in nominal units.

A.2 Generalization

We apply the envelope theorem of Milgrom and Segal (2002) to allow a positive mass of workers to have several initially optimal choices. In this subsection and Section A.3, we measure compensation in units of real consumption. If maximized expected utility rises by G, a transfer of G units of real consumption under initial conditions raises utility by G at every choice and makes the worker as well off as under terminal conditions. And removing the same amount under terminal conditions restores initial utility. These transfers leave the ranking of choices unchanged, so the argument applies even when several choices are initially optimal. In order to transfer the same G units of real consumption across different regions requires nominal payments that vary with the price index. With initial ties across regions with different prices, fixed nominal transfers can change choices, and nominal equivalent and compensating variation can differ even to first order.

Let $t \geq 0$ parametrize a change in the economic environment, with $t = 0$ denoting the initial conditions. Write expected utility from region-sector choice $a = (k,s)$ as $f^\omega(a,t)$. Assume payoffs are measurable across workers, which ensures that individual welfare changes

are also measurable and can be integrated over the population. For each worker and choice, assume expected utility is continuously differentiable in t, and denote its derivative by $g^\omega(a,t) \equiv \partial f^\omega(a,t)/\partial t$. We obtain this derivative by differentiating expected utility as in Section A.1, now evaluating all terms at choice (k,s) and time t. For some $\bar{t} > 0$, assume each worker's expected utility changes at a bounded rate for every choice: $|g^\omega(a,t)| \leq M^\omega$ for all a and $t \in [0,\bar{t}]$. The bound may differ across workers, but assume $\int_{\Omega_r} M^\omega d\omega < \infty$. These assumptions ensure that aggregate welfare changes are finite and that aggregating individual first-order approximations gives a first-order approximation of aggregate welfare. Let a_0^ω denote the worker's actual initial choice, and let A_0^ω denote the set of choices that maximize the worker's expected utility under initial conditions.

Let $V^\omega(t) \equiv \max_a f^\omega(a,t)$ denote worker ω's maximized expected utility. Its change equals the required real consumption transfer. We define the consumption-equivalent welfare measure for the set Ω_r of initial residents of region r as

$$\widetilde{W}_r(t) \equiv \frac{1}{RI_r} \int_{\Omega_r} [V^\omega(t) - V^\omega(0)]\, d\omega,$$

where RI_r denotes these workers' initial total expected real income. Let $W_r(t)$ denote the main text's aggregate nominal equivalent variation divided by initial nominal income for the change from 0 to t.

Define the expression evaluated at workers' actual initial choices as

$$\widehat{W}_r(t) \equiv \frac{t}{RI_r} \int_{\Omega_r} g^\omega(a_0^\omega, 0) d\omega.$$

This is the right-hand side of equation (1) with changes written as t times their initial derivatives. Under the baseline uniqueness assumption, $W_r(t) = \widehat{W}_r(t) + o(t)$.

Proposition 1. *As $t \downarrow 0$,*

$$\widetilde{W}_r(t) = \frac{t}{RI_r} \int_{\Omega_r} \max_{a \in A_0^\omega} g^\omega(a,0) d\omega + o(t).$$

Equivalently,

$$\widetilde{W}_r(t) = \widehat{W}_r(t) + tT_r + o(t),$$

where

$$T_r \equiv \frac{1}{RI_r} \int_{\Omega_r} \left[\max_{a \in A_0^\omega} g^\omega(a,0) - g^\omega(a_0^\omega, 0) \right] d\omega \geq 0.$$

$\widehat{W}_r(t)$ approximates $\widetilde{W}_r(t)$ with an error of $o(t)$ if and only if $T_r = 0$. This occurs if and only if, for almost every worker, the actual initial choice gives the largest payoff derivative among the worker's

initially optimal choices. In particular, $T_r = 0$ if either the mass of initially indifferent workers is zero or all initially tied choices have the same payoff derivative for almost every worker.

When almost every worker has a unique initial choice, $\widetilde{W}_r(t) = W_r(t) + o(t)$.

Proof. We show that the difference between $\widetilde{W}_r(t)$ and the proposed approximation is $o(t)$. By the definition of $\widetilde{W}_r(t)$,

$$\begin{aligned}
&\frac{1}{t}\left|\widetilde{W}_r(t) - \frac{t}{RI_r}\int_{\Omega_r} \max_{a\in A_0^\omega} g^\omega(a,0)d\omega\right| \\
&\quad = \frac{1}{RI_r}\left|\int_{\Omega_r}\left[\frac{V^\omega(t)-V^\omega(0)}{t} - \max_{a\in A_0^\omega} g^\omega(a,0)\right]d\omega\right| \\
&\quad \leq \frac{1}{RI_r}\int_{\Omega_r}\left|\frac{V^\omega(t)-V^\omega(0)}{t} - \max_{a\in A_0^\omega} g^\omega(a,0)\right|d\omega.
\end{aligned}$$

With finitely many choices and continuously differentiable payoffs, Corollary 4 of Milgrom and Segal (2002) implies that the integrand in the final line tends to zero for each worker. It remains to show that the integral over the population also tends to zero. The bound on payoff derivatives gives $|V^\omega(t) - V^\omega(0)|/t \leq M^\omega$ and $\left|\max_{a\in A_0^\omega} g^\omega(a,0)\right| \leq M^\omega$. Thus, the integrand is bounded above by $2M^\omega$. Since this bound has a finite population integral, the integral in the final line also tends to zero. The aggregate approximation error divided by t therefore tends to zero, establishing the first formula in the proposition.

As in Section A.1, multiplying $g^\omega(a_0^\omega, 0)$ by t, integrating over initial residents, and dividing by RI_r gives $\widehat{W}_r(t)$. Subtracting $\widehat{W}_r(t)$ from $\widetilde{W}_r(t)$ gives

$$\begin{aligned}
\widetilde{W}_r(t) - \widehat{W}_r(t) &= \frac{t}{RI_r}\int_{\Omega_r}\left[\max_{a\in A_0^\omega} g^\omega(a,0) - g^\omega(a_0^\omega,0)\right]d\omega + o(t) \\
&= tT_r + o(t).
\end{aligned}$$

Since $a_0^\omega \in A_0^\omega$, the integrand defining T_r is nonnegative. Hence $T_r \geq 0$, and $\widehat{W}_r(t)$ approximates $\widetilde{W}_r(t)$ with an error of $o(t)$ if and only if $T_r = 0$. A nonnegative integrand has integral zero if and only if it is zero for almost every worker, which gives the stated condition on actual initial choices. A unique initial choice, or equal derivatives across all initially optimal choices, makes this integrand zero. □

The consumption-equivalent measure $\widetilde{W}_r(t)$ and normalized nominal equivalent variation $W_r(t)$ can differ for finite changes or, even to first order, when a positive mass of workers is initially indifferent between regions with different prices.

Proposition 1 shows the role of our assumption that the mass of initially indifferent

workers is zero. Among choices that initially give a worker the same expected utility, a small change can make one better than the others. To first order, the worker's consumption-equivalent welfare change equals the largest change in expected utility among these initially optimal choices. Evaluating the change at the worker's actual initial choice can therefore understate a welfare gain or overstate a welfare loss, where $\widehat{W}_r(t)$'s aggregate first-order understatement of $\widetilde{W}_r(t)$ is measured by tT_r. This understatement is zero ($tT_r = 0$) if the initial choice experiences the largest first-order change among the choices the worker is initially indifferent between for almost every worker. One sufficient condition for that to be achieved is that almost no workers are indifferent between initial optimal choices; this is the assumption we impose in the paper.

Measuring T_r directly would require identifying each worker's initially optimal choices and the payoff changes associated with each, including choices not taken. Corollary 1 avoids this requirement by evaluating the effects of the change at each worker's observed terminal choice. This gives the correct first-order consumption-equivalent welfare change even when a positive mass of workers is initially indifferent.

Corollary 1. *Let $a^\omega(t) \in \arg\max_a f^\omega(a,t)$ denote worker ω's terminal choice. Assume that, for each t, the set of workers making each choice is measurable. Under the payoff assumptions stated before Proposition 1, allowing arbitrary initial ties,*

$$\widetilde{W}_r(t) = \frac{t}{RI_r}\int_{\Omega_r} g^\omega(a^\omega(t),t)d\omega + o(t).$$

Proof. By the definition of $\widetilde{W}_r(t)$,

$$\begin{aligned}
&\frac{1}{t}\left|\widetilde{W}_r(t) - \frac{t}{RI_r}\int_{\Omega_r} g^\omega(a^\omega(t),t)d\omega\right| \\
&\quad= \frac{1}{RI_r}\left|\int_{\Omega_r}\left[\frac{V^\omega(t)-V^\omega(0)}{t} - g^\omega(a^\omega(t),t)\right]d\omega\right| \\
&\quad\le \frac{1}{RI_r}\int_{\Omega_r}\left|\frac{V^\omega(t)-V^\omega(0)}{t} - g^\omega(a^\omega(t),t)\right|d\omega.
\end{aligned}$$

For each worker, every choice outside A_0^ω gives strictly lower initial expected utility than the choices in A_0^ω. Because there are finitely many choices and payoffs are continuous, none of these initially inferior choices can become optimal for sufficiently small t. Hence $a^\omega(t) \in A_0^\omega$, so both initial and terminal maximized utility can be evaluated at the terminal choice. Holding that choice fixed when integrating its payoff derivative gives

$$\frac{V^\omega(t)-V^\omega(0)}{t} - g^\omega(a^\omega(t),t) = \frac{1}{t}\int_0^t \left[g^\omega(a^\omega(t),\tau) - g^\omega(a^\omega(t),t)\right]d\tau.$$

For each fixed choice, continuity of its payoff derivative implies that the right-hand side tends to zero. Since there are finitely many choices, this remains true whichever terminal choice is selected as t changes.

It remains to show that the integral over workers also tends to zero. As in the proof of Proposition 1, $|V^\omega(t) - V^\omega(0)|/t \leq M^\omega$, and the terminal payoff derivative is also bounded in absolute value by M^ω. The integrand in the final line of the first display is therefore bounded by $2M^\omega$. Since this bound has a finite population integral, that integral tends to zero. The aggregate approximation error is therefore $o(t)$, proving the corollary. □

Formulas using terminal choices. To express Corollary 1 in the notation of equation (1), let Ω^T_{rks} contain initial residents of r who choose (k,s) in the terminal equilibrium, and let Ω^T_{rk} be their union over sectors. Superscript T denotes terminal values. Define $RI^{e,T}_{rks}$, $RI^{u,T}_{rks}$, $\eta^{e,T}_{rks}$, and $\eta^{u,T}_{rks}$ as in equation (1), using these groups and terminal conditions. Thus, $RI^{e,T}_{rks}$ and $RI^{u,T}_{rks}$ denote terminal expected total real income from employment and unemployment, respectively, for initial residents of r who choose (k,s) in the terminal equilibrium. Write $RI^{u,T}_{rk} \equiv \sum_{s=1}^{S} RI^{u,T}_{rks}$ and let $\Phi^T_{rkm} \equiv \int_{\Omega^T_{rk}} \frac{d\phi^\omega_m(z^T_{km})}{dz^T_{km}} d\omega$ sum their terminal marginal valuations of characteristic m. Replacing derivatives by small observed changes gives

$$\begin{aligned}
\widetilde{W}_r \approx & \sum_k \sum_{s=0}^{S} \frac{RI^{e,T}_{rks}}{RI_r} d\log\left(\frac{w_{ks}}{P_k}\right) + \sum_k \frac{RI^{u,T}_{rk}}{RI_r} d\log\left(\frac{w_{ku}}{P_k}\right) \\
& + \sum_k \sum_{s=1}^{S} \frac{RI^{e,T}_{rks} - \frac{E^T_{ks}}{1-E^T_{ks}} RI^{u,T}_{rks}}{RI_r} d\log E_{ks} \\
& + \sum_k \sum_{s=1}^{S} \frac{\eta^{e,T}_{rks} - \frac{E^T_{ks}}{1-E^T_{ks}} \eta^{u,T}_{rks}}{RI_r} d\log E_{ks} + \sum_k \sum_{m=1}^{M} \frac{\Phi^T_{rkm}}{RI_r} dz_{km}.
\end{aligned} \tag{A.1}$$

The denominator RI_r remains initial real income. Terminal choices determine which changes and weights enter each worker's welfare calculation. Each change is still measured within a fixed destination region or region-sector pair. For wages, this means holding both the worker and the pair fixed when comparing earnings across dates. For a worker who moves to (k,s), we compare the earnings she would receive in (k,s) under initial and terminal conditions. We do not compare her initial earnings in the pair she leaves with her terminal earnings in (k,s).

For the analogue of equation (2), impose its restriction on terminal choices: $E^T_{ks} = E^T_k$ for $s \geq 1$. Define terminal income totals and average incomes in destination nominal units, and define η^T_{rk} and $\overline{\phi}^T_{m,rk}$ as in equation (2), for these workers. The indices rk identify their initial

and terminal regions, and $N_{rk}^T \equiv |\Omega_{rk}^T|$. Then

$$\widetilde{W}_r \approx \sum_k \frac{P_r^0}{P_k^T} \left[\sum_{j \in \mathcal{J}} \frac{I_{rkj}^T}{I_r} d \log \left(\frac{w_{kj}}{P_k} \right) + (1 + \eta_{rk}^T) \frac{I_{rk}^{e,T}}{I_r} \left[1 - \frac{\overline{I}_{rk}^{u,T}}{\overline{I}_{rk}^{e,T}} \right] d \log E_k + \sum_{m=1}^{M} \frac{N_{rk}^T \overline{\phi}_{m,rk}^T}{I_r} dz_{km} \right]. \tag{A.2}$$

Here P_r^0 is the initial price index in r, P_k^T is the terminal price index in k, and $I_r = P_r^0 R I_r$ is initial nominal income.

Measurement. Implementing the terminal formula requires observing initial residents' terminal allocations across CZ, sector, and labor force status. For a given initial CZ, we would have to construct terminal income totals and population counts across all destinations, together with unemployment income, non-pecuniary unemployment costs, and valuations of the regional characteristics for those workers.

Wage growth would still be measured within fixed CZ-sector pairs, using stayers under our maintained assumption of common proportional productivity changes. Each group's terminal weights would be applied to changes in wages, prices, employment probabilities, and the regional characteristics in its destination. The price ratios additionally require comparable price levels across locations and dates; the inflation series used in our application supply only changes over time. Welfare would still be assigned to the initial CZ and divided by its initial income.

If $\widehat{W}_r(t)$ approximates $\widetilde{W}_r(t)$ with an error of $o(t)$ under Proposition 1, no change in its construction is needed.

A.3 Finite changes

Our first-order welfare formula omits higher-order effects for finite changes. A fully specified structural model can account for these effects, although the resulting welfare estimates depend on its additional assumptions and parameters. Here, we ask how observed data can help assess sensitivity to our approximation without specifying such a model.

Specifically, we derive exact lower and upper bounds on the consumption-equivalent welfare measure $\widetilde{W}_r(t)$ defined in Section A.2. These bounds apply to finite changes, whether or not a positive mass of workers is initially indifferent.

We then show that the initial-choice formula in equation (1) and the terminal-choice formula in equation (A.1) provide first-order approximations to the lower and upper bounds, respectively. Because the approximation errors can go in either direction, the formulas themselves need not bound $\widetilde{W}_r(t)$ for finite changes.

Maintain the notation of Section A.2. Let a_0^ω denote worker ω's initial choice and $a^\omega(t)$ an optimal choice under terminal conditions. Define

$$B_r^0(t) \equiv \frac{1}{RI_r}\int_{\Omega_r}\left[f^\omega(a_0^\omega,t)-f^\omega(a_0^\omega,0)\right]d\omega,$$
$$B_r^T(t) \equiv \frac{1}{RI_r}\int_{\Omega_r}\left[f^\omega(a^\omega(t),t)-f^\omega(a^\omega(t),0)\right]d\omega. \tag{A.3}$$

The first expression evaluates the exact change in expected utility at workers' initial choices; the second evaluates it at their terminal choices. Both follow the initial residents of r and use their initial real income as the denominator.

Proposition 2. *For any finite change in the economic environment,*

$$B_r^0(t) \le \widetilde{W}_r(t) \le B_r^T(t),$$

allowing for arbitrary initial and terminal ties.

Proof. Worker ω's required real consumption transfer equals the change in maximized expected utility,

$$V^\omega(t)-V^\omega(0) = f^\omega(a^\omega(t),t)-f^\omega(a_0^\omega,0).$$

Optimality at the terminal and initial conditions implies, respectively,

$$f^\omega(a_0^\omega,t) \le f^\omega(a^\omega(t),t),$$
$$f^\omega(a^\omega(t),0) \le f^\omega(a_0^\omega,0).$$

Consequently,

$$f^\omega(a_0^\omega,t)-f^\omega(a_0^\omega,0) \le V^\omega(t)-V^\omega(0) \le f^\omega(a^\omega(t),t)-f^\omega(a^\omega(t),0).$$

Integrating over Ω_r and dividing by RI_r gives the result. □

These bounds apply to the consumption-equivalent measure $\widetilde{W}_r(t)$. For finite changes, this measure can differ from nominal equivalent variation relative to initial nominal income, $W_r(t)$, so even the exact bounds need not contain $W_r(t)$.

The difference between $\widetilde{W}_r(t)$ and the lower bound is the gain from changing choices under terminal conditions. The difference between the upper bound and $\widetilde{W}_r(t)$ is the loss from taking the terminal choices under initial conditions. For a worker whose choice does not change, both bounds equal that worker's consumption-equivalent welfare change.

We next show that the initial-choice formula provides a first-order approximation to the exact lower bound, while the terminal-choice formula provides a first-order approximation

to the exact upper bound. Under the regularity conditions of Section A.2, Taylor expansion at the initial and terminal conditions gives

$$B_r^0(t) = \frac{t}{RI_r}\int_{\Omega_r} g^\omega(a_0^\omega, 0)d\omega + o(t),$$
$$B_r^T(t) = \frac{t}{RI_r}\int_{\Omega_r} g^\omega(a^\omega(t), t)d\omega + o(t). \tag{A.4}$$

Because payoff derivatives are continuous and there are finitely many choices, each worker's approximation error divided by t tends to zero even if the terminal choice changes with t. The integrable bound on derivatives ensures that this also holds after aggregating across workers.

The first term in the first line is the initial-choice formula in equation (1). The first term in the second line is the terminal-choice formula in equation (A.1). Under the additional restrictions used in equation (2), their simplified expressions are equations (2) and (A.2), respectively. Thus, the formulas in the preceding two subsections approximate the exact lower and upper bounds to first order.

For finite changes, the exact bounds may be close or far apart, even when almost every worker has a unique initial choice. Their first-order approximations can be constructed from observed allocations and outcomes, together with the required monetary valuations, without specifying a full structural model. Comparing these approximations provides one way of judging the sensitivity of our first-order approach to measuring welfare changes. Because the formulas themselves approximate the bounds, however, the difference between them does not provide a guaranteed bound on approximation error.

A.4 Risk aversion

This subsection allows workers to be risk averse. Our treatment of unemployment risk is related to Baily (1978) and Chetty (2006), who show how consumption differences between employment and unemployment and risk aversion inform the welfare benefits of unemployment insurance. We first derive the general formula, then show how to implement it using the data and monetary valuations in our application.

Replace linear utility from consumption with the constant relative risk aversion (CRRA) utility function

$$u(c) = \begin{cases} \dfrac{c^{1-\gamma}-1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1, \end{cases}$$

where $\gamma \geq 0$ is the common coefficient of relative risk aversion. Expected utility from choos-

ing (r,s) is

$$\mathbb{E}[U^\omega(r,s)] = E_{rs}\left[u(C^\omega_{rs}) + \eta^\omega_{rs}\right] + (1-E_{rs})\left[u(C^\omega_{ru}) + \eta^\omega_{ru}\right] + \sum_{m=1}^{M} \phi^\omega_m(z_{rm}).$$

The functions η^ω_{rj} and ϕ^ω_m now express amenities in utility units. Their monetary valuations are defined below. As in the main text, job amenities are fixed, productivity changes are proportional and common within each region-job pair, and almost every worker has a unique initial optimal choice. Consumption is positive, and $E_{rs} > 0$ for $s \geq 1$. Non-participants receive C^ω_{r0} with certainty.

For workers initially applying to (r,s), assume

$$C^\omega_{ru} = \rho_{rs} C^\omega_{rs}, \qquad \omega \in \Omega_{rs}, \quad s \geq 1, \tag{A.5}$$

where $0 < \rho_{rs} < 1$. This restriction permits arbitrary differences in consumption levels across workers, but requires their proportional consumption loss from unemployment to be common within each initial group. The restriction applies in the initial equilibrium; replacement ratios can subsequently change. When $0 < E_{rs} < 1$, the initial income totals defined in Section 2.2 identify the common replacement ratio:

$$\rho_{rs} = \frac{E_{rs}}{1-E_{rs}} \frac{RI^u_{rs}}{RI^e_{rs}}.$$

Thus, if employment and unemployment income totals are observed for each initial group, they identify its common replacement ratio.

We retain the welfare definition in the main text. Equivalent variation, Y^ω, is a fixed annual nominal transfer under initial conditions and before the employment lottery that makes the worker indifferent between the initial and terminal environments. The transfer is independent of the worker's choice and employment outcome. Welfare for initial residents of r is $W^\gamma_r \equiv (P_r RI_r)^{-1} \int_{\Omega_r} Y^\omega d\omega$, with the initial population and income denominator held fixed. Equivalent and compensating variation coincide to a first-order approximation under the assumption of a unique initial choice for almost every worker.[1]

For an initial applicant to (r,s), the marginal expected utility of a nominal transfer is Λ^ω_{rs}/P_r, where

$$\Lambda^\omega_{rs} \equiv E_{rs} u'(C^\omega_{rs}) + (1-E_{rs}) u'(C^\omega_{ru}) = (C^\omega_{rs})^{-\gamma} A_{rs},$$

[1]This extension maintains the uniqueness assumption and does not directly extend the result allowing a positive mass of initially indifferent workers. As in Section A.2, we impose a bound that allows individual first-order approximations to be aggregated. For all sufficiently small changes, each worker's equivalent and compensating variations are bounded in absolute value by a worker-specific amount times the size of the change. These worker-specific amounts have a finite population integral.

and where

$$A_{rs} \equiv E_{rs} + (1 - E_{rs})\rho_{rs}^{-\gamma}$$

is expected marginal utility of consumption relative to marginal utility of consumption when employed. For non-participants, define $\Lambda_{r0}^{\omega} \equiv (C_{r0}^{\omega})^{-\gamma}$. We also define

$$L_\gamma(\rho) \equiv \begin{cases} \dfrac{1-\rho^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ -\log\rho, & \gamma = 1. \end{cases}$$

This function gives the consumption utility gap between employment and unemployment, with $u(C_{rs}^{\omega}) - u(C_{ru}^{\omega}) = (C_{rs}^{\omega})^{1-\gamma} L_\gamma(\rho_{rs})$.

To express non-pecuniary effects in monetary units, define

$$H_{rs}^{\gamma} \equiv \int_{\Omega_{rs}} \frac{\eta_{rs}^{\omega} - \eta_{ru}^{\omega}}{\Lambda_{rs}^{\omega}} d\omega,$$
$$\Phi_{rm}^{\gamma} \equiv \sum_{s=0}^{S} \int_{\Omega_{rs}} \frac{1}{\Lambda_{rs}^{\omega}} \frac{d\phi_m^{\omega}(z_{rm})}{dz_{rm}} d\omega.$$

The non-pecuniary welfare effect of a change dE_{rs} is $H_{rs}^{\gamma} dE_{rs}$, in annual real consumption units. Likewise, Φ_{rm}^{γ} sums initial residents' marginal willingness to pay for characteristic m.

Welfare formula. Under these assumptions, equation (1) becomes

$$\begin{aligned} W_r^{\gamma} \approx\ & \frac{RI_{r0}^{e}}{RI_r} d\log\left(\frac{w_{r0}}{P_r}\right) + \sum_{s=1}^{S} \frac{RI_{rs}^{e}}{RI_r A_{rs}} d\log\left(\frac{w_{rs}}{P_r}\right) \\ & + \sum_{s=1}^{S} \frac{\rho_{rs}^{-\gamma} RI_{rs}^{u}}{RI_r A_{rs}} d\log\left(\frac{w_{ru}}{P_r}\right) \\ & + \sum_{s=1}^{S} \left[\frac{RI_{rs}^{e}}{RI_r} \frac{L_\gamma(\rho_{rs})}{A_{rs}} + \frac{E_{rs} H_{rs}^{\gamma}}{RI_r}\right] d\log E_{rs} \\ & + \sum_{m=1}^{M} \frac{\Phi_{rm}^{\gamma}}{RI_r} dz_{rm}. \end{aligned} \tag{A.6}$$

All quantities entering the weights and monetary valuations are evaluated initially. Setting $\gamma = 0$ gives $A_{rs} = 1$ and $L_0(\rho_{rs}) = 1 - \rho_{rs}$, yielding equation (1).

Proof. The envelope theorem evaluates the first-order change in maximized expected utility at the worker's initial choice. Dividing this change by $\Lambda_{rs}^{\omega}/P_r$ converts it into nominal

equivalent variation. Thus, for a worker initially applying to (r,s),

$$\frac{Y^\omega}{P_r} \approx \frac{1}{\Lambda^\omega_{rs}}\Bigg[E_{rs}u'(C^\omega_{rs})dC^\omega_{rs} + (1-E_{rs})u'(C^\omega_{ru})dC^\omega_{ru} + \left(u(C^\omega_{rs}) - u(C^\omega_{ru}) + \eta^\omega_{rs} - \eta^\omega_{ru}\right)dE_{rs} + \sum_{m=1}^{M}\frac{d\phi^\omega_m(z_{rm})}{dz_{rm}}dz_{rm}\Bigg].$$

Write $g_{rj} \equiv d\log(w_{rj}/P_r)$. Common proportional productivity changes imply $dC^\omega_{rj} = C^\omega_{rj}g_{rj}$. Substituting CRRA utility, the replacement ratio in equation (A.5), and $dC^\omega_{rj} = C^\omega_{rj}g_{rj}$ into this expression gives

$$\frac{Y^\omega}{P_r} \approx \frac{C^\omega_{rs}}{A_{rs}}\left[E_{rs}g_{rs} + (1-E_{rs})\rho_{rs}^{1-\gamma}g_{ru} + L_\gamma(\rho_{rs})dE_{rs}\right] + \frac{\eta^\omega_{rs}-\eta^\omega_{ru}}{\Lambda^\omega_{rs}}dE_{rs} + \sum_{m=1}^{M}\frac{1}{\Lambda^\omega_{rs}}\frac{d\phi^\omega_m(z_{rm})}{dz_{rm}}dz_{rm}.$$

The coefficient multiplying C^ω_{rs} is common across initial applicants to (r,s). Aggregating this term across workers therefore requires only total income. In particular, $RI^e_{rs} = E_{rs}\int_{\Omega_{rs}} C^\omega_{rs}d\omega$ and $RI^u_{rs} = (1-E_{rs})\rho_{rs}\int_{\Omega_{rs}} C^\omega_{rs}d\omega$. For non-participants, the consumption contribution is $C^\omega_{r0}g_{r0}$. Integrating Y^ω/P_r over initial residents, using $dE_{rs} = E_{rs}d\log E_{rs}$, and dividing by RI_r gives equation (A.6). □

Implementation with regional data. Under risk aversion, valuing income losses requires assumptions about how they affect consumption. This extension retains our static environment and identifies consumption with current real income. Saving and borrowing can reduce the consumption loss associated with unemployment, so income replacement rates need not equal consumption replacement rates when workers can make these choices. Assumptions about insurance also matter. For example, Rodriguez-Clare et al. (2026) allow workers within each region-sector group to share unemployment risk through transfers from employed to unemployed workers.

Our application observes unemployment and replacement rates at the regional level. We therefore retain $E_{rs} = E_r$ and additionally impose $\rho_{rs} = \rho_r$ for all initial applicants in region r. Under these restrictions, the individual consumption replacement ratio equals the ratio of average unemployment income to average employment income, $\rho_r = \bar{I}^u_r/\bar{I}^e_r$, using the model's identification of consumption with real income. Define $A_r \equiv E_r + (1-E_r)\rho_r^{-\gamma}$.

We express the monetary non-pecuniary cost of unemployment relative to the income

loss as

$$\eta_r \equiv \frac{\sum_{s=1}^{S} H_{rs}^{\gamma}}{(1-\rho_r)\sum_{s=1}^{S}\int_{\Omega_{rs}} C_{rs}^{\omega} d\omega}.$$

This definition preserves the monetary interpretation of η_r in the main text. Individual non-pecuniary valuations can differ arbitrarily. Similarly, let $\overline{\phi}_{m,r} \equiv P_r \Phi_{rm}^{\gamma} / N_r$ be the average marginal valuation of characteristic m in nominal income units. Expressing incomes in nominal units as in equation (2), we obtain

$$\begin{aligned} W_r^{\gamma} \approx & \frac{I_{r0}}{I_r} d\log\left(\frac{w_{r0}}{P_r}\right) + \frac{1}{A_r}\sum_{s=1}^{S}\frac{I_{rs}}{I_r} d\log\left(\frac{w_{rs}}{P_r}\right) + \frac{\rho_r^{-\gamma}}{A_r}\frac{I_{ru}}{I_r} d\log\left(\frac{w_{ru}}{P_r}\right) \\ & + \frac{I_r^e}{I_r}\left[\frac{L_\gamma(\rho_r)}{A_r} + \eta_r(1-\rho_r)\right] d\log E_r + \sum_{m=1}^{M}\frac{N_r\overline{\phi}_{m,r}}{I_r} dz_{rm}. \end{aligned} \tag{A.7}$$

The application holds real returns to unemployment and home production fixed and specializes regional characteristics to life expectancy, $z_{r1} = \ell_r$, with average marginal valuation $\overline{\phi}_{\ell,r} \equiv \overline{\phi}_{1,r}$. Equation (A.7) therefore becomes

$$\begin{aligned} W_r^{\gamma} \approx & \underbrace{\frac{1}{A_r}\sum_{s=1}^{S}\frac{I_{rs}}{I_r} d\log\left(\frac{w_{rs}}{P_r}\right)}_{\text{real wage}} \\ & + \underbrace{\frac{I_r^e}{I_r}\left[\frac{L_\gamma(\rho_r)}{A_r} + \eta_r(1-\rho_r)\right] d\log E_r}_{\text{unemployment}} + \underbrace{\frac{N_r\overline{\phi}_{\ell,r}}{I_r} d\ell_r}_{\text{life expectancy}}. \end{aligned} \tag{A.8}$$

Relative to the main text, risk aversion multiplies the real wage component by $1/A_r$ and the pecuniary unemployment component by $L_\gamma(\rho_r)/[A_r(1-\rho_r)]$.

To value the non-pecuniary cost, we assume that the payment needed to compensate a worker for unemployment is k times that worker's income loss, with the same compensation multiple $k \geq 0$ for all regions. The payment is received if unemployed. Holding the worker's region-sector choice fixed, this means

$$u\left([\rho_r + k(1-\rho_r)]C_{rs}^{\omega}\right) + \eta_{ru}^{\omega} = u(C_{rs}^{\omega}) + \eta_{rs}^{\omega}.$$

This equality determines the non-pecuniary utility difference. Dividing it by the marginal

expected utility of a real transfer, Λ_{rs}^{ω}, and using the definition of η_r gives

$$\eta_r = \frac{1}{A_r(1-\rho_r)} \begin{cases} \dfrac{[\rho_r + k(1-\rho_r)]^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ \log[\rho_r + k(1-\rho_r)], & \gamma = 1. \end{cases} \tag{A.9}$$

Because k and ρ_r are common within the region, each worker's monetary non-pecuniary cost is proportional to that worker's employed income. Individual income levels therefore cancel from η_r. When $\gamma = 0$, $\eta_r = k - 1$, so $k = 3$ recovers the main text's $\eta_r = 2$. Holding k fixed as γ changes generally changes η_r. Given γ and k, this calibration uses the existing regional income and employment data, without requiring the full distribution of incomes or preferences.

For life expectancy, we retain the HHS value of a statistical life used in Section C.2.5. We treat it as the initial average willingness to pay for a small reduction in annual mortality risk and apply it to the same mortality changes. The monetary life expectancy contribution therefore remains unchanged as γ varies. The total welfare outcome and its components can then be used in the same regressions as in the main text.

B Data sources

Here, we provide details about the raw data sources used in our analysis and our approach to data cleaning and preparation. This includes information on how regions and industries in each dataset are converted to the commuting zones (CZs) and 6-digit 1997 NAICS codes (if applicable), any filtering criteria applied (e.g., age restrictions, sample selection), and methods for handling missing values or outliers.

Our analysis primarily relies on the following datasets:

- **IPUMS-USA (Census and American Community Survey):** We downloaded the following samples from the Integrated Public Use Microdata Series (IPUMS-USA):
  - **Census 1990 5% State sample**
  - **Census 2000 5% sample**
  - **2006–2008 American Community Survey (ACS) 3-year sample (3% density):** This sample combines the 1% ACS samples for 2006, 2007, and 2008 into a single dataset.

  These datasets provide comprehensive demographic, labor force, and income information at the individual level. The cleaning and preparation of these data involved several steps, closely following the methodology outlined in Autor et al. (2013).

- **Variable Selection:** We retained key variables including year, state, county group, PUMA, person weight, metropolitan area codes, group quarters status, employment status, industry codes (NAICS and 1990 Census industry), weeks worked, usual hours worked, wage income, total income, and other income.
- **Weeks and Hours Worked Imputation:** For the ACS sample, weeks worked and hours worked are reported in intervals rather than as continuous values. We converted these to continuous variables by assigning the midpoint of each interval.
- **Missing Value Treatment:** We recoded values marked as missing or not applicable as missing, following standard IPUMS coding conventions.
- **Income Topcoding:** Following Autor et al. (2013), wage income, total income, and other income were topcoded at 1.5 times their respective topcode thresholds for each survey year.
- **Hourly Wage Calculation and Truncation:** We calculated hourly wages as wage income divided by the product of usual hours worked per week and weeks worked per year. We top-coded hourly wages using year-specific maximum values derived from top-coded wage income and by assuming standard work hours (35 hours/week for 50 weeks/year). We set hourly wages of zero to be missing. We bottom-coded hourly wages at the 1st percentile for each year to address outliers.
- **Sample Restrictions:** We applied the following criteria to define our sample:
  - We retained individuals aged 18 to 65 years.
  - We excluded individuals in institutional or military group quarters.
  - We excluded military personnel, identified using detailed employment status codes and 1990 Census industry codes for military-related industries.
  - We dropped observations with missing education or missing/undefined industry codes, including those who last worked 5+ years ago or never worked.
- **Commuting Zone (CZ) Aggregation:** Individual-level data were aggregated to the commuting zone (CZ) level using IPUMS-provided geographic identifiers and the crosswalks available on David Dorn's website. Person weights were applied during aggregation to ensure representativeness.

- **IPUMS Time Use (American Heritage Time Use Study - AHTUS):** The American Heritage Time Use Study (AHTUS) data was downloaded from IPUMS Time Use. This dataset is used to measure time allocation for non-market work, specifically to derive the ratio of time use in non-market work by those not in the labor force relative to time use in market work by full-time employees.

- **Sample Selection:** We selected individuals who were neither employed nor unemployed. This ensures that our sample contains only respondents who are out of the labor force. We further restricted the sample to individuals aged 18 to 65 years.
- **Non-Market Work Measurement:** We constructed total time spent on non-market work by summing time allocated to undifferentiated domestic activities and civic/religious activities.
- **Home Production Share Calculation:**
  - The numerator for the home production replacement rate is the average time use of non-participants on non-market work. Based on the 1993 AHTUS data, the average minutes spent on non-market work by those out of the labor force on a given day is 220.63 minutes.
  - The denominator is the average minutes spent on work by a full-time employee. This is approximated as 342.86 minutes (60 minutes $\times$ 40 hours a week / 7 days a week).
  - The average share of hours worked by those in home production is approximately 0.64 (= 220.63/342.86).
- **Weighting and Aggregation:** We weighted individual observations using record weights provided by AHTUS. We then aggregated these weighted averages to the national level, by sample year, to obtain the required time use ratios.

- **Confidential Longitudinal Employer-Household Dynamics (LEHD) Data:** We utilize confidential data from the Longitudinal Employer-Household Dynamics (LEHD) program. This dataset provides detailed information on worker earnings and employment spells, crucial for measuring changes in wages per efficiency unit by holding worker composition fixed.

  The LEHD data are derived from quarterly earnings reports provided by employers to state unemployment insurance (UI) agencies. They are assembled by the U.S. Census Bureau. The core records contain total quarterly earnings paid by each employer to each worker, along with a small set of worker and establishment characteristics, including industry and location. Coverage extends to roughly 95% of private-sector employment and to state and local government workers, but excludes federal employees, members of the armed forces, and the self-employed.

  For our main analysis, we extract all available LEHD earnings records for two reference quarters: 1999Q2 and 2007Q2. We restrict the sample in each quarter to individuals aged 18-57 in 1999Q2, ensuring workers are at most 65 years old in 2007Q2. We also

restrict the sample to those employed in a single job (dropping those with multiple jobs within the quarter) to ensure a unique industry assignment. In all cases, industry and location are defined from the worker's unique employer in the quarter. For our main measures of wages per efficiency unit, we retain only workers who appear in both periods and who are employed in the same commuting zone and industry in both quarters. Observations with missing industry codes and/or commuting zone identifiers are dropped. The resulting sample represents full-quarter employment spells for a consistent set of workers over time, excluding movers across industries and/or commuting zones, part-time workers with multiple concurrent jobs, and other atypical employment arrangements. All earnings measures correspond to total quarterly earnings for these workers. Due to state participation patterns in LEHD, our measures cover commuting zones in 34 states. Industry and geographic codes are harmonized to 6-digit 1997 NAICS and commuting zones using Census-provided crosswalks.

For the falsification exercise, we apply the same restrictions but instead focus on the period 1992Q2 to 1999Q2, which results in fewer observations due to declining state coverage in earlier years. When constructing the alternative earnings outcomes in Panel A of Table 2 of the main text, we keep the age and single-job restrictions. Column 2 drops the requirement that workers be in the same commuting zone and industry in both quarters, and column 3 also drops the requirement that they be employed in 2007Q2. Column 2 fixes the set of workers employed in the commuting zone in 1999Q2 who have positive earnings in 2007Q2 in any commuting zone or industry. For this set, we compute mean log earnings in each quarter and take the difference, which equals the average log earnings change across these workers. Column 3 fixes the set of all workers employed in the commuting zone in 1999Q2, including those with zero earnings in 2007Q2. For this fixed set of workers, we compute mean earnings in levels in each quarter, allowing for zero earnings in 2007Q2, and then take the difference in the logs of these means. Both columns deflate earnings by the national CPI-U-RS, scale changes by 7/8 to a seven-year equivalent, and weight all workers equally within a commuting zone. They include workers in every commuting zone-industry pair, including pairs dropped from the real wage component because fewer than 20 workers are employed in the pair in both quarters. The set of commuting zones is the same across the three columns. Unlike the real wage component, they are not multiplied by employed workers' initial income share, I_r^e / I_r. Since non-employment enters the real wage component with zero change, the component is roughly I_r^e / I_r (0.96 on average) times an income-weighted average of sector wage changes. This makes both the coefficient and the implied 90-10 difference in column 1 smaller in absolute value relative to columns 2 and 3. Without it, the gap between column 1 and columns 2 and 3 would

be slightly larger.

- **Confidential Longitudinal Business Database (LBD) Data:** The Longitudinal Business Database (LBD) provides comprehensive information on firm and establishment characteristics. This data is used for measuring total number of employed workers and the total wagebill within each sector (I_{rs}), as well as to construct commuting zone-level trade shocks.
  - **Data Sources and Years:** We utilize LBD data for the years 1980, 1990, and 2000. The raw LBD data includes county-level information for each year and records industries using 2002 6-digit NAICS codes. Establishments with zero employment are excluded from the sample.
  - **Geographic Concordance:** We convert the county-level information in the LBD to commuting zones (CZs) by applying crosswalks available on David Dorn's website.
  - **Industry Concordance:** To align with our defined sector classification, we crosswalk the 2002 6-digit NAICS codes to 1997 6-digit NAICS codes using a Census crosswalk file.[2]
- **Commuting Zone Controls:** Commuting zone-level control variables used in our regression analysis—including the manufacturing share, college share, foreign-born share, female employment share, routine occupation share, and average offshorability—are taken directly from Autor et al. (2013).
- **Price Index Data (BLS CPI and State-Level CPI):** We use two primary sources for price index data.
  - **National CPI (CPI-U-RS):** The Consumer Price Index research series for all Urban Consumers using current methods (CPI-U-RS) was downloaded from the Bureau of Labor Statistics (BLS) website.
    - **Cleaning and Processing:** The series was rebased to January 1990. Annual average CPI values were then calculated for the years 1990, 2000, and 2007. Log differences were computed for the periods 1990–2000 and 2000–2007. The 1990–2000 difference was scaled by a factor of 7/10 to make it comparable to a 7-year equivalent change, consistent with our benchmark period.

[2] This crosswalk involves handling instances where the mapping is not one-to-one. Specifically, among 1354 unique matches between 1997 NAICS and 2002 NAICS codes, approximately 300 are not one-to-one matches (i.e., a single 2002 NAICS code maps to multiple 1997 NAICS codes, or vice-versa). In such cases, we equally assign the industries.

- **State-Level Price Index:** We utilize state-level price indices available from Hazell et al. (2022), which provides regional inflation data. This data was sourced from their supplementary materials.
    - **Cleaning and Processing:** The raw quarterly state-level CPI data was imported and quarterly inflation was calculated. Similar to the national CPI, the state-level series were rebased to 1990Q1. Annual average state-level CPI values were then computed for 1990, 2000, and 2007. Log differences were calculated for the 1990–2000 and 2000–2007 periods, with the 1990–2000 difference also scaled to a 7-year equivalent.
- **Missing Data Handling and Merging:** State-level price indices were merged with commuting zone (CZ) data. To ensure comprehensive coverage across all CZs, for states where the state-level CPI data was unavailable (17 states, including the District of Columbia), the national CPI-U-RS difference was used as a substitute. This imputation strategy ensures that all CZs have a corresponding price index measure for both common and regional inflation analyses. Note that the simple average of state-level inflation across all available CZs is similar to the national inflation for both periods: 0.163 for state-level inflation versus 0.172 for national inflation in 1990–2000, and 0.182 for state-level inflation versus 0.186 for national inflation in 2000–2007.

- **Department of Labor (Unemployment Replacement Rate):** The state-level unemployment replacement rate reports were directly downloaded from the Department of Labor website: `https://oui.doleta.gov/unemploy/ui_replacement_rates.asp`. This data is used to construct the unemployment income component.
    - **Data Selection:** Specifically, we utilize "Replacement Ratio 2," which is defined as the Ratio of: Weighted Average Weekly Benefit Amount (WBA) / Weighted Average (Normal Hourly Wage x 40 Hrs.). The WBA represents the claimant's weekly benefit amount.
    - **Coverage and Use:** The UI replacement rate covers a four-quarter moving average ending with a selected quarter-ending date. The data is available from 1997, and the 1997 unemployment replacement rate is used for both our main and falsification analyses.
    - **Source Details:** This report uses the Unemployment Insurance Benefit Accuracy Measurement (BAM) data, stored in the UI database.

- **NTR Gap Data:** Sector-level NTR gaps are taken from Pierce and Schott (2016). The

NTR gap is defined as the difference between the non-NTR tariff rate and the NTR tariff rate for each industry, measured as of 1990.

C Construction of welfare measures

This section provides detailed documentation on the construction of our welfare measure, W_r, for both the baseline analysis period (2000–2007) and the pre-trend analysis period (1992–2000). We first describe the welfare equation and how each component is measured, and then discuss the specific years and scaling adjustments used for each period.

C.1 Welfare equation

For the application, we specialize equation (2) of the main text to life expectancy, setting $M = 1$ and $z_{r1} = \ell_r$. Writing $\overline{\phi}_{\ell,r} \equiv \overline{\phi}_{1,r}$, the formula becomes:

$$W_r \approx \underbrace{\sum_{j \in \mathcal{J}} \frac{I_{rj}}{I_r} d \log \left(\frac{w_{rj}}{P_r} \right)}_{\text{real wage}} + \underbrace{(1 + \eta_r) \frac{I_r^e}{I_r} \left[1 - \frac{\overline{I}_r^u}{\overline{I}_r^e} \right] d \log E_r}_{\text{unemployment}} + \underbrace{\frac{N_r \overline{\phi}_{\ell,r}}{I_r} d\ell_r}_{\text{life expectancy}} \tag{C.10}$$

where $\mathcal{J}$ denotes the set of all jobs (each sector of employment, labor force non-participation, and unemployment). The first term represents the real wage component of welfare, the second term represents the unemployment component, and the third term represents the life expectancy component. For the real wage component, we measure CZ $\times$ 6-digit NAICS sector income (I_{rs}) from the Longitudinal Business Database (LBD) and wage changes ($d \log w_{rs}$) from the Longitudinal Employer-Household Dynamics (LEHD). For the unemployment component and for constructing CZ-level aggregates, we use IPUMS Census data.

C.2 Measurement of welfare components

We now describe how each component of equation (C.10) is measured in detail. The specific years and data sources used for each analysis period are described in Section C.3.

C.2.1 Income shares

The real wage component of equation (2) of the main text requires measuring income shares I_{rj}/I_r for each job j. We describe the construction of each income component below.

Total Income (I_r). Total income in region r is the sum of employment, unemployment, and non-participation income:

$$I_r = I_r^e + I_r^u + I_{r0} \tag{C.11}$$

Employment Income (I_r^e). Total employment income in region r is measured as the sum of total wagebill across all sectors:

$$I_r^e = \sum_{s=1}^{S} I_{rs} \tag{C.12}$$

where I_{rs} is the total wage bill within CZ-sector pair (r, s), measured from the LBD (for the real wage component) or IPUMS Census data (for the unemployment component).

Non-Participation Income (I_{r0}). Total income of non-participants in region r is constructed as:

$$I_{r0} = \bar{I}_{r0} \times N_{r0} \tag{C.13}$$

where $\bar{I}_{r0}$ is the average income of non-participants and N_{r0} is the number of non-participants. This can be rewritten as:

$$I_{r0} = \frac{\bar{I}_{r0}}{\bar{I}_r^e} \times \frac{N_{r0}}{N_r^e} \times I_r^e \tag{C.14}$$

where $\bar{I}_{r0}/\bar{I}_r^e$ is the home production replacement rate and N_{r0}/N_r^e is the ratio of non-participants to employed workers. We construct each component as follows. First, the home production replacement rate is constructed as the product of two terms:

$$\frac{\bar{I}_{r0}}{\bar{I}_r^e} = \left(\frac{\text{time use, non-market work by NILF}}{\text{time use, work by employed}}\right) \times \left(\frac{\text{avg wage, private household sector}}{\text{avg wage, all sectors}}\right) \tag{C.15}$$

where NILF refers to those not in the labor force. The first term captures the relative time devoted to home production, while the second term values this time using average wages in the private household sector (as a proxy for the market value of home production activities) relative to average wages across all sectors.

- **Time use ratio:** We measure this ratio using the American Heritage Time Use Study (AHTUS) data. The numerator is the average time spent on non-market work by those not in the labor force; based on the 1993 AHTUS data, this equals 220.63 minutes per day. The denominator is the average time spent on work by a full-time employee, approximately 342.86 minutes per day (40 hours per week divided by 7 days). The resulting ratio is approximately 0.64.

- **Wage ratio:** We measure this ratio using IPUMS Census data, where the private house-

hold sector is defined using the 1990 Census industry code for "Private households." The ratio is computed as the average wage per worker in the private household sector divided by the average wage per worker across all sectors, measured at the state level. We measure this ratio at the state level rather than the CZ level because some CZ-level estimates yield implausible values due to small sample sizes; the state-level ratio is then applied uniformly to all CZs within each state.

Second, the ratio of non-participants to employed workers can be expressed as:

$$\frac{N_{r0}}{N_r^e} = \frac{L_r - M_r}{E_r M_r} \tag{C.16}$$

where L_r is the total population aged 18–65, M_r is the labor force (employed plus unemployed), and $E_r = N_r^e / M_r$ is the employment rate. Note that $L_r - M_r$ equals the number of non-participants and $E_r M_r = N_r^e$.

- **Population ratio:** The ratio $(L_r - M_r)/(E_r M_r)$ is measured by commuting zone from IPUMS Census data.

Unemployment Income (I_r^u). Total income of unemployed workers in region r is constructed as:

$$I_r^u = \frac{\overline{I}_r^u}{\overline{I}_r^e} \times \frac{N_r^u}{N_r^e} \times I_r^e \tag{C.17}$$

where $\overline{I}_r^u / \overline{I}_r^e$ is the unemployment replacement rate and $N_r^u / N_r^e = (1 - E_r)/E_r$ is the ratio of unemployed to employed workers.

- **Unemployment replacement rate ($\overline{I}_r^u / \overline{I}_r^e$):** State-specific unemployment replacement rate from the Department of Labor. We use "Replacement Ratio 2," defined as the ratio of weighted average weekly benefit amount to weighted average (normal hourly wage $\times$ 40 hours).
- **Population ratio:** The ratio $(1 - E_r)/E_r$ is measured by commuting zone from IPUMS Census data.

C.2.2 Changes in productivity-adjusted wages per efficiency unit

Sectoral Wages ($d \log w_{rs}$ for $s \geq 1$). For each sector $s \geq 1$, we measure $d \log w_{rs}$ using LEHD panel data. We construct a sample of workers who are employed in commuting zone r and sector s in both the initial and terminal quarters. On this sample, we compute

$$\widehat{d \log w_{rs}} = \frac{1}{N_{rs}} \sum_{i=1}^{N_{rs}} \left(\log w_{i,t'} - \log w_{i,t} \right). \tag{C.18}$$

Here t and t' denote the initial and terminal quarters, N_{rs} is the number of stayers in the pair, and $w_{i,t}$ is worker i's total quarterly earnings. This holds worker composition fixed.

To state the condition for unbiasedness, let $w_{i,t'}(r,s)$ denote the terminal earnings that an initial worker in (r,s) would receive if employed in that pair, including workers who leave. Write

$$\log w_{i,t'}(r,s) - \log w_{i,t} = \mu_{rs} + u_{i,rs}, \tag{C.19}$$

where μ_{rs} is average log earnings growth if all initial workers had stayed, and $u_{i,rs}$ is a worker's deviation from this average, so $\mathbb{E}[u_{i,rs} \mid r,s] = 0$. Let $D_{i,rs} = 1$ indicate that the worker is employed in the same pair in both periods. Among these workers, the left-hand side is observed, and equation (C.18) is the OLS estimate of the pair-specific intercept in equation (C.19). It is unbiased for μ_{rs} if

$$\mathbb{E}[u_{i,rs} \mid D_{i,rs} = 1, r, s] = 0. \tag{C.20}$$

Thus, stayers must have the same average earnings growth that all initial workers would have experienced had they stayed. The relevant comparison uses the earnings workers who leave would have received in their original pair, which are unobserved. A positive average residual among stayers produces upward bias; a negative average produces downward bias.

The condition in equation (C.20) makes the stayer estimator unbiased for average log earnings growth among all initial workers in the pair. Welfare aggregation requires an average weighted by initial earnings, because the same proportional earnings change produces a larger dollar gain or loss for a worker with higher initial earnings. The two averages coincide if earnings growth is uncorrelated with initial earnings within the pair.

Under the common proportional productivity changes assumed in the main text, both requirements hold: $\mu_{rs} = d \log w_{rs}$ and $u_{i,rs} = 0$ for every initial worker. Productivity levels may differ arbitrarily across workers and region-job pairs, and staying may depend on these levels. Taking log differences removes these fixed differences. The restriction concerns how productivity evolves over time: within each pair, workers have the same proportional productivity growth. This common growth can be positive or negative.

If these conditions fail, errors in wage growth enter the CZ welfare outcome through the initial income weights. Such errors bias PNTR's estimated differential welfare effect only if the resulting welfare error is correlated with exposure after accounting for the controls and regression weights.

Unemployment and Non-Participation ($j \in \{u, 0\}$). As described in the main text, we choose changes in nominal returns to unemployment and non-participation such that real

wage changes are zero:

$$d \log \left(\frac{w_{rj}}{P_r} \right) = 0 \quad \text{for } j \in \{u, 0\} \tag{C.21}$$

which implies $d \log w_{rj} = d \log P_r$ for these jobs.

Handling Missing Wage Data. In constructing the real wage component of equation (2) of the main text, there are some CZ-sector pairs for which changes in productivity-adjusted wages per efficiency unit are not defined (due to insufficient sample size, i.e., fewer than 20 workers employed in the same CZ-sector pair in both the initial and terminal quarters), yet the weight I_{rs}/I_r in the weighted average is not zero. For these CZ-sector pairs, we set the weights to zero and proportionally rescale the remaining weights, including those on unemployment and home production, so that they sum to one.

C.2.3 Regional price indices

We measure region-specific inflation, $d \log P_r$, using state-level CPI data from Hazell et al. (2022). For states without state-level CPI data (17 states including the District of Columbia), we use the national CPI-U-RS as a substitute. As a robustness check, we also apply the national CPI-U-RS for all CZs.

C.2.4 Unemployment component

The unemployment component of equation (2) of the main text requires the following elements:

Change in Employment Rate ($d \log E_r$). The employment rate E_r is measured as the ratio of employed population to labor force (one minus the unemployment rate) from IPUMS Census data.

Non-Pecuniary Cost of Unemployment (η_r). The parameter η_r represents the non-pecuniary welfare loss of unemployment relative to the pecuniary welfare loss. The simpler welfare measure sets $\eta_r = 0$. The more complete measure sets $\eta_r = 2$, which implies multiplying the unemployment component of welfare by a factor of 3 (i.e., $1 + \eta_r = 3$).

C.2.5 Life expectancy component

The simpler welfare measure sets this component to zero. Because mortality is not observed in our data, the more complete measure uses the estimates in Pierce and Schott (2020). They find that an interquartile-range increase in county exposure to PNTR raises annual drug-overdose mortality by approximately 2 to 3 deaths per 100,000 population; we use the mid-point, 2.5. This range summarizes their annual estimates after PNTR over 2001-2013. They

report that their results are robust to ending the sample in 2007, the end of our analysis period. They find no relationship between PNTR and mortality from suicide or alcohol-related liver disease, or from the 16 major categories of internal causes of death, such as cancer and diseases of the respiratory system. Drug overdoses therefore account for the mortality effect they estimate.

We value the increase in annual mortality risk using the value of a statistical life, V, and divide by annual income per capita, $\bar{I}$. We use the same ratio $V/\bar{I}$ for every CZ. We set V to 14.1 million dollars, the central value adopted by the U.S. Department of Health and Human Services for 2026, expressed in 2025 dollars (U.S. Department of Health and Human Services, 2026). We set $\bar{I}$ to 76,328 dollars, U.S. personal income per capita in 2025 in the data used for this calculation (U.S. Bureau of Economic Analysis, 2026). Equation (C.10) divides welfare changes by I_r, which includes employment income, unemployment income, and the value of home production for the initial workers. Summing this constructed income across CZs and dividing by total population gives approximately 71 percent of U.S. personal income per capita in 1999. We apply this ratio to our income denominator, replacing $\bar{I}$ with $0.71\,\bar{I}$. The resulting ratio is approximately 260.

The interquartile range of exposure in Pierce and Schott (2020) is 8.3 percentage points. Their exposure measure averages NTR gaps over all industries, assigning zero to industries whose output is not subject to tariffs. Our measure, NTRG_r, averages over manufacturing sectors with a defined gap. We approximate the conversion between these measures using the income-weighted mean manufacturing employment share across our CZs in 1980, $\overline{m} = 0.23$. With exposure expressed as a fraction, the implied welfare loss per unit of NTRG_r is

$$\kappa \equiv \frac{V}{0.71\,\bar{I}} \frac{2.5/100{,}000}{0.083}\,\overline{m} \approx 0.018.$$

We therefore assign CZ r the mortality contribution

$$W_r^{\text{mort}} = -\kappa\,\text{NTRG}_r.$$

This imputation includes only the differential mortality effect implied by the published estimate. It omits mortality changes common to all CZs and differential changes due to other shocks. With direct mortality data, we could instead value observed changes, including both of these omitted contributions.

Adding W_r^{mort} to the welfare outcome and applying the same regression, with the same sample, controls, and weights, lowers the coefficient on NTRG_r by κ. Thus, the coefficient changes from -0.296 in column 2 to -0.314 in column 3 of Panel B of Table 2 of the main text. The regression residuals are unchanged, so the reported standard error is unchanged

when κ is treated as fixed.

The gap in $NTRG_r$ between CZs at the 90th and 10th percentiles of exposure is about 7.7 percentage points. Under our approximation, this implies about 0.5 additional deaths per 100,000 and a differential welfare loss associated with granting China PNTR of $\kappa \times 0.077$, or approximately 0.14 percent of annual income. This is an annual welfare loss; we do not accumulate it over the years following PNTR. Using mortality responses of 2 to 3 deaths per 100,000 gives losses of 0.11 to 0.17 percent. Holding the mortality response at 2.5 and using the low and high HHS statistical life values, 6.6 and 21.5 million dollars, gives losses of 0.06 to 0.21 percent.

We also consider an alternative income denominator. As a sensitivity exercise, we use $\overline{I}$ without this adjustment. This lowers κ to approximately 0.013 and the mortality loss at the 90th-10th percentile exposure gap to 0.10 percent of annual income. Mortality then accounts for approximately 4 percent of the differential welfare effect, compared with 6 percent under our baseline normalization.

C.3 Analysis periods

C.3.1 Baseline period (2000–2007)

For our main analysis, we measure welfare changes over the baseline period 2000-2007. Note that the specific years used for each component vary slightly depending on data availability and timing considerations, as detailed below.

Income Shares. We measure income shares using year 2000 data from the following sources:

- **Employment income (I_{rs}):** Longitudinal Business Database (LBD), year 2000 (for the real wage component); IPUMS Census 2000 (for the unemployment component).
- **Wage ratio for home production:** IPUMS Census 2000, measured at the state level.
- **Time use ratio for home production:** American Heritage Time Use Study (AHTUS) 1993. This ratio (approximately 0.64) is time-invariant in our analysis.
- **Population counts** (for non-participation and unemployment): IPUMS Census 2000, measured at the CZ level.
- **Unemployment replacement rate:** Department of Labor, 1997 (the earliest year available).

We use 2000 because the LBD provides the most reliable establishment-level data in decennial Census years due to enhanced coverage and accuracy during these benchmark periods.

The IPUMS data similarly benefits from the larger sample sizes available in decennial Census years.

Changes in Productivity-Adjusted Wages per Efficiency Unit. We measure $d \log w_{rs}$ using LEHD data from 1999Q2 to 2007Q2. We use 1999Q2 as the initial quarter to ensure that any anticipation effects associated with the Congressional vote on PNTR in October 2000 and China's accession to the WTO in December 2001 are captured in the baseline period rather than the pre-trend period.

Because the LEHD wage changes span 8 years (from 1999Q2 to 2007Q2), we scale these changes by 7/8 to construct a 7-year equivalent measure that is comparable to the nominal 7-year baseline period (2000–2007).

Employment Rate. We measure $d \log E_r$ using the 2000 Decennial Census and the combined 2006–2008 ACS samples. The use of multiple ACS years increases sample sizes at the commuting zone level.

Price Index. We measure $d \log P_r$ using state-level CPI data from Hazell et al. (2022) for 2000–2007.

C.3.2 Pre-trend period (1992–2000)

For our falsification exercise, we measure welfare changes over the pre-trend period 1992-2000 to check that our baseline results capture the effects of granting China PNTR rather than the continuation of pre-existing trends. As with the baseline period, the specific years used for each component vary depending on data availability.

Income Shares. We measure income shares using year 1990 data from the following sources:

- **Employment income (I_{rs}):** Longitudinal Business Database (LBD), year 1990 (for the real wage component); IPUMS Census 1990 (for the unemployment component).
- **Wage ratio for home production:** IPUMS Census 1990, measured at the state level.
- **Time use ratio for home production:** American Heritage Time Use Study (AHTUS) 1993 (same as baseline).
- **Population counts** (for non-participation and unemployment): IPUMS Census 1990, measured at the CZ level.
- **Unemployment replacement rate:** Department of Labor, 1997 (same as baseline; the earliest year available).

We use 1990 because this is the only decennial Census year available for the pre-trend period, and the LBD and IPUMS data provide reliable coverage in this benchmark year comparable to that of the 2000 data used in our baseline analysis.

Changes in Productivity-Adjusted Wages per Efficiency Unit. We measure $d \log w_{rs}$ using LEHD data from 1992Q2 to 1999Q2. We use 1999Q2 as the terminal quarter (rather than 2000Q2) to ensure that any anticipation effects of China's WTO accession are captured by the baseline period and not the pre-trend period. The 7-year span from 1992Q2 to 1999Q2 requires no scaling adjustment since it already matches the 7-year baseline equivalent.

State coverage in the LEHD is substantially more limited in this earlier period due to the gradual expansion of state participation in the LEHD program over time. Whereas the baseline period covers 34 states and approximately 600 commuting zones, the pre-trend period covers only 6 states and approximately 80 commuting zones.

Employment Rate. We measure $d \log E_r$ using the 1990 and 2000 Decennial Censuses. Because this spans 10 years rather than 7, we scale the change in $\log E_r$ by $7/10$ to construct a 7-year equivalent measure comparable to our baseline period.

Price Index. We measure $d \log P_r$ using state-level CPI data from Hazell et al. (2022) for 1990–2000. As with the employment rate, the 10-year price change is scaled by $7/10$ to obtain a 7-year equivalent.

D Additional results

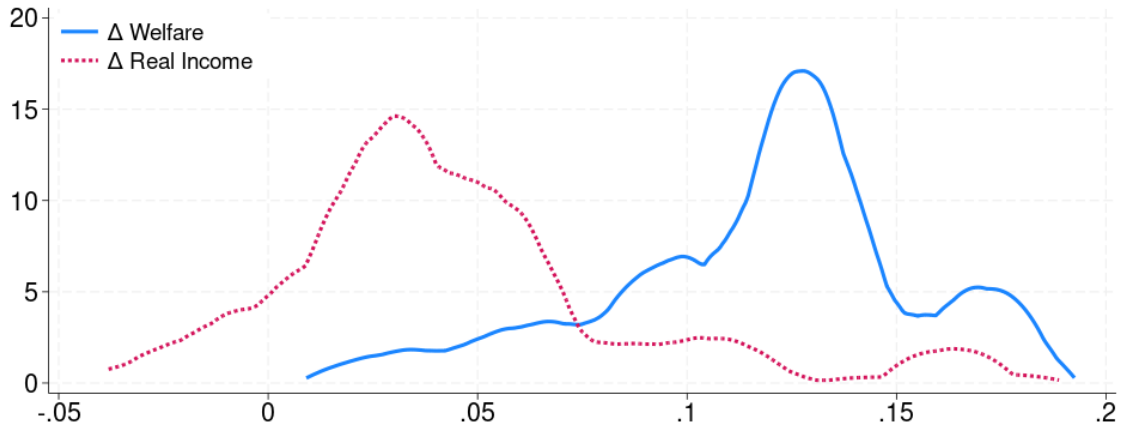
D.1 Observed welfare changes across CZs

Figure D.1 reports the distribution of welfare changes under our simpler measure between 2000 and 2007, weighting CZs by initial employment income.[3] The median is a gain equivalent to 12.4% of initial annual income, with gains of 6% and 17.2% at the 10th and 90th percentiles. These changes reflect all shocks over the period and omit non-pecuniary unemployment costs and mortality.

The figure also reports changes in average real wage income among workers employed in each CZ in each period. The median increase is approximately 4%. This outcome follows a changing workforce: entry, exit, and moves across sectors and regions affect its composition. Our welfare statistic concerns a fixed set of initial residents.

The age composition also differs. Average wage income in the figure holds the age range fixed at 18-64 in both years, while the workers represented by the welfare statistic age over

[3] U.S. Census Bureau disclosure restrictions prohibit releasing statistics for individual CZs, so we report the distribution rather than a CZ-level map.



Welfare changes (solid blue) and changes in average real wage income (dotted red), weighted by CZ employment income in 2000.

Figure D.1: The distribution of changes in welfare across CZs, 2000-2007

time. When we instead compare workers aged 18-57 initially with workers aged 25-64 in the terminal year, mean real wage income growth and mean welfare growth are similar. This accounts for the difference in the locations of the two distributions, without making the underlying outcomes equivalent.

D.2 Regional prices and the welfare components

Table D.1 shows how regional prices and the remaining welfare components enter the PNTR estimates. With a national price index, the real wage component implies a differential effect equivalent to 2.1% of initial income between CZs at the 90th and 10th percentiles of exposure. Using state-level indices where available reduces this effect to 1.7%, since more exposed regions experience relative price declines. Adding pecuniary unemployment costs gives the differential of 1.9% in our simpler measure. The final two columns add non-pecuniary unemployment costs and life expectancy, as in Panel B of Table 2 of the main text.

D.3 The distribution of exposure across CZs

Figure D.2 displays the distribution of NTRG_r across CZs.

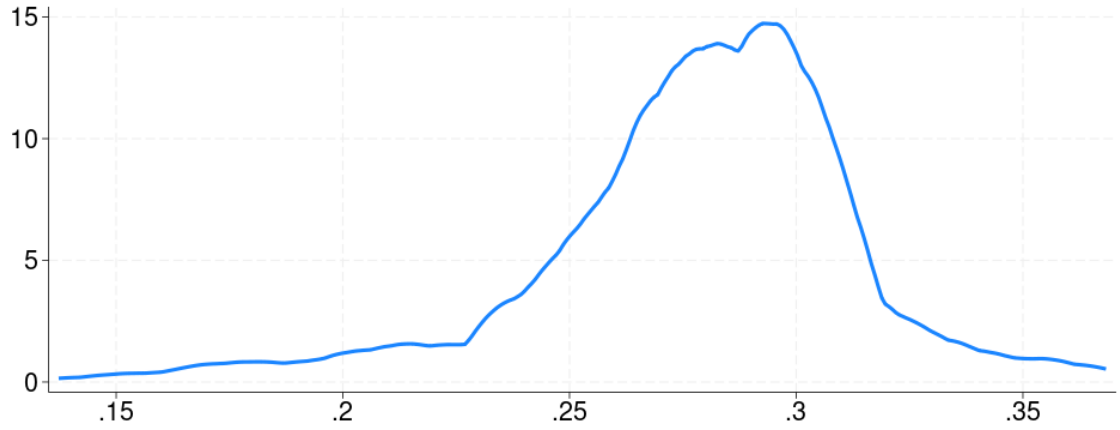
D.4 Sector-level and region-level exposure

What explains the strong correlation between the real wage component of regional welfare and the regional measure of exposure to granting China PNTR? One possibility is that the combination of changes in real wages per efficiency unit and productivity at the region-sector level, $d \log w_{rs}$, is similar across regions within a sector, but that some regions have

	real wage component (1)	+ regional price index (2)	+ unemployment component, $\eta_r = 0$ (3)	+ unemployment component, $\eta_r = 2$ (4)	+ life expectancy (5)
$NTRG_r$	-0.266***	-0.224**	-0.248***	-0.296***	-0.314***
	(0.085)	(0.093)	(0.096)	(0.104)	(0.104)
$\mathbb{E}[W_r : NTRG_{p90} - NTRG_{p10}]$	-.021	-.017	-.019	-.023	-.024
Baseline			✓		
Observations	600	600	600	600	600

All columns use the specification in column 5 of Table 1 of the main text. Column 1 uses the real wage component with a national price index; column 2 substitutes state-level indices where available. Column 3 adds pecuniary unemployment costs. Column 4 sets $\eta_r = 2$, adding non-pecuniary unemployment costs. Column 5 adds the mortality contribution described in Section C.2.5. Standard errors are clustered by state and condition on the valuations and published mortality estimate. Regressions are weighted by CZ employment income in 2000.

Table D.1: Contributions of prices, unemployment, and life expectancy to the differential welfare effect.



The distribution of $NTRG_r$ across CZs, weighted by CZ income among employed workers in the year 2000.

Figure D.2: The distribution of $NTRG_r$ across CZs.

higher shares of their incomes earned in more exposed sectors, where changes in $d \log w_{rs}$ are less positive. Another possibility is that changes in the combination of wages per efficiency unit and productivity at the region-sector level are less positive in more exposed regions within sectors. We investigate these possibilities by regressing $d \log w_{rs}$ separately and jointly on the sectoral and regional measures of the NTR Gap. Results are reported in Table D.2. We find that the regional measure of the NTR Gap is strongly and negatively correlated with $d \log w_{rs}$ whereas the sectoral measure of the NTR Gap is not. This suggests that the real wage component of regional welfare falls more in regions with a higher regional measure of exposure to granting China PNTR because the combination of changes in real wages per efficiency unit and productivity falls more within sectors in these regions. This conclusion is related to results in Pierce et al. (2024). Also using the LEHD, they find that granting China PNTR lowered incomes of workers initially employed in more exposed

regions, but within a region, not more so in sectors that are more directly exposed. Our results in Table D.2 differ from Pierce et al. (2024) because we measure changes in the combination of wages per efficiency unit and productivity (on the sample of workers who start and end employed in the same region-sector pair), which is what our theory requires for welfare analysis, whereas they measure changes in incomes for workers who start employed in a given region and sector, which is their focus.

	$d \log w_{rs}$		
	(1)	(2)	(3)
Sectoral NTR Gap$_s$	-0.056		-0.004
	(0.042)		(0.033)
Regional NTRG$_r$		-0.573***	-0.569***
		(0.146)	(0.137)
Baseline regional controls		✓	✓
$\mathbb{E}[d \log w_{rs} : \text{NTRG}_{p90(r)}\text{-NTRG}_{p10(r)}]$		-.055	-.054
$\mathbb{E}[d \log w_{rs} : \text{NTR Gap}_{p90(s)}\text{-NTR Gap}_{p10(s)}]$	-.015		-.0012
Observations	26500	26500	26500

Each column presents results of regressing $d \log w_{rs}$, measured between 2000 and 2007, on either NTR Gap$_s$ (in column 1), or on NTRG$_r$ and the baseline set of controls included in column 5 of Table 1 of the main text (in column 2), or both (in column 3). Regressions are weighted by CZ-sector employment and standard errors are clustered by state. The number of CZ-sector observations is an approximation (because of reporting requirements).

Table D.2: Changes in wages per efficiency unit, sector-level NTR Gap$_s$, and CZ-level NTRG$_r$.

D.5 Treating non-participants as unemployed

In our baseline, we assume that workers choose optimally between participating in the labor force and home production (labor force non-participation). Hence, we measure the probability of successfully finding a job using a regional version of the U-3 unemployment rate, which is the official unemployment rate reported by the U.S. Bureau of Labor Statistics. To check the sensitivity of our empirical results to the assumption that those not participating in the labor force are on their labor supply curves, in columns 3 and 4 of Table D.3 we instead assume that workers who report being out of the labor force are off their labor supply curves, like the unemployed. In column 3 we remeasure the unemployment replacement rate, using an average of unemployment benefits and home production income, weighting by the shares of these populations. In column 4, we instead use our baseline unemployment replacement rate. Both results are very similar to our baseline results. Column 5 combines the adjustment of column 3 with the non-pecuniary cost of unemployment, $\eta_r = 2$, of column 2; the differential welfare impact of granting China PNTR is slightly larger in this specification,

reflecting the combined effects of both adjustments.

	W_r				
	(1)	(2)	(3)	(4)	(5)
$NTRG_r$	-0.248***	-0.296***	-0.254**	-0.248**	-0.314***
	(0.096)	(0.104)	(0.100)	(0.100)	(0.120)
$\mathbb{E}[W_r : NTRG_{p90}\text{-}NTRG_{p10}]$	-.019	-.023	-.02	-.019	-.024
Baseline	✓				
Non-pecuniary cost of unemployment		✓			✓
NILF + unemployment, adjusted replacement			✓		✓
NILF + unemployment, baseline replacement				✓	
Observations	600	600	600	600	600

Column 1 replicates Column 5 in Table 1 of the main text. The remaining columns estimate the same specification but with slight adjustments to the measurement of welfare. Column 2 incorporates a non-pecuniary cost of unemployment into the calculation of welfare, setting $\eta_r = 2$. Column 3 treats non-participants as unemployed and adjusts the unemployment replacement rate to include income both among the unemployed and those out of the labor force; NILF refers to not-in-the-labor-force. Column 4 treats non-participants as unemployed using the baseline measure of the unemployment replacement rate. Column 5 combines the adjustments of columns 2 and 3.

Table D.3: Sensitivity to the treatment of labor force non-participation.

D.6 Falsification

To verify that our results capture the effects of exposure to granting China PNTR rather than the continuation of pre-existing trends, we regress changes in welfare between 1992 and 2000 on future exposure to granting China PNTR, on the smaller sample of 6 states and approximately 80 CZs covered by the LEHD in that period, compared to 34 states and approximately 600 CZs in our baseline period.[4]

	Welfare change		
	2000-2007		1992-2000
	(1)	(2)	(3)
$NTRG_r$	-0.248***	-0.314**	0.302
	(0.096)	(0.141)	(0.180)
$\mathbb{E}[W_r : NTRG_{p90}\text{-}NTRG_{p10}]$	-.019	-.024	.023
Baseline sample	✓		
Placebo sample		✓	✓
Observations	600	80	80

Column 1 replicates Column 5 in Table 1 of the main text. Column 2 displays results of estimating this specification on the smaller sample of CZs observed in the LEHD over the 1992–2000 period. Finally, column 3 replaces the 2000-2007 change in welfare with the value between 1992 and 2000 and estimates on the smaller sample of column 2. The number of CZ observations is an approximation (because of reporting requirements).

Table D.4: Falsification: changes in welfare between 1992 and 2000.

[4]Section C.3 describes the measurement of welfare changes in the pre-period.

Column 2 of Table D.4 shows that the decrease in state and CZ coverage does not materially affect our baseline results over the 2000-2007 period. Column 3 provides little evidence consistent with a continuation of differential pre-trends across more and less exposed CZs: the estimated coefficient is insignificantly different from zero, and welfare was, if anything, rising in more exposed CZs before China was granted PNTR.

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